

MODERN OPTICS PROJECT LABORATORY



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Modern Optics Project Laboratory

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Licensing

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CHAPTER OVERVIEW

1: Safety, Polarization States of Optical Waves, Dielectric Reflection and Geometric Optics

[1.1: Pre-lab Exercises](#)

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1.1: Pre-lab Exercises

Safety, Polarization States of Optical Waves, Dielectric Reflection and Geometric Optics

The Pre-Lab Exercises must be completed BEFORE entering the Lab. In your lab notebook record data, explain phenomena you observe, and answer the questions asked. Remember to answer all questions in your lab notebook in a neat and orderly fashion. No data are to be taken on these laboratory sheets. Tables provided herein are simply examples of how to record data into your laboratory notebooks. Expect the in-lab portion of this exercise to take about 3 hours.

PL 1.1 - Safety

Read the entire Laboratory and laser Safety Packet that was handed out in the classroom. Be prepared to answer questions about high-voltage and laser safety before arriving at the MOL.

PL 1.2 - Get Prepared to Start the Laboratory Exercises

Read the entire laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. Write them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade.

PL 1.3 - Telescope Design

For this exercise, you are encouraged to work in groups of 2-3 people. In this exercise, you are to design a telescope given the optical components listed in Table I. Your telescope must be a refractor (lenses-only). The entire telescope assembly must not be any longer than the 2 feet (the shorter, the better). Your telescope will need to contain at least 2 optical elements, but no more than 4. The telescope must be a terrestrial telescope (meaning that the image as seen by a viewer must be erect). Your total system magnification must be at least 4x and at most 40x at an imaging distance of either 15 meters or adjusted to infinity (a collimating telescope – this is the most common form of telescope, and is preferred for simplicity and robustness of design). Be sure to use the ABCD matrix approach to simplify the design of your telescope. You should include all of your system specifications in the pre-lab write-up – including any computer code, hand-calculations or other materials used in the design. Be sure to specify the focal lengths, total system magnification as well as the locations of the lenses, the object and the image. Remember, there is no one ‘correct’ way to design your telescope. The Lab staff will be very sad if everyone designs the same telescope. Additional points will be awarded for ingenuity and thoughtfulness in design. Also, be sure not to use too many elements; each additional element in your system will lead to decreased contrast and decreased resolution. “Ingenuity and thoughtfulness” means designing smart, and minimizing complexity.

TABLE 1. Optical elements available for building your telescope:

Size	Type	Focal Length	Quantity
2"	Bi-Convex	200 mm	1
2"	Bi-Convex	100 mm	2
2"	Bi-Convex	75 mm	1
1"	Bi-Convex	200 mm	1
1"	Bi-Convex	150 mm	1
1"	Bi-Convex	100 mm	1
1"	Bi-Convex	75 mm	1
1"	Bi-Convex	50 mm	1
1"	Bi-Convex	35 mm	2
1"	Bi-Convex	25.4 mm	2
1"	Bi-Concave	-100 mm	1

1"	Bi-Concave	-75 mm	1
1"	Bi-Concave	-50 mm	1
1"	Bi-Concave	-25 mm	1

Hints: Remember to use larger lenses at the input of your system (the side facing the world), and smaller lenses close to the eye. Also, before you start designing, determine which element in the ABCD matrix must equal zero for angular magnification to occur.

PL 1.4 - Extra Credit (Not Required)

Design a zoom-optic system with the components listed above. (With a zoom-optic system, you can adjust the magnification of your telescope by moving a central lens while keeping the outer most lenses at *approximately* the same location – basically the same setup as is used in an SLR camera zoom lens.)

"Two brothers bought a cattle ranch and named it "Focus." When their father asked why they chose that name, they replied: "It's the place where the sons raise meat." -- Prof. W. B. Pietenpol, Physics Department, University of Colorado, Boulder, Colorado

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1.2: In-lab Exercises

Safety, Polarization States of Optical Waves, Dielectric Reflection and Geometric Optics

1.1 Linear Polarizer

1. As was discussed in lecture, light reflected off the waxed floor tiles is predominantly polarized perpendicular to the plane of incidence. Holding the given sheet of linear polarizer close to your eye, use such reflections to determine the axis of the polarizer. Convince the T.A. that you have identified the polarizer axis.
2. Explain using electric dipole arguments and with the aid of diagrams why the light reflected from the floor wax is predominantly polarized perpendicular to the plane of incidence.

1.2 Two-Polarizer Diagnostic System

The setup you are given consists of a randomly polarized (unpolarized) laser beam of intensity I_0 , a pair of polarizers, and a screen arranged as illustrated in Figure 2. The second polarizer is often called an analyzer. Its axis is generally oriented perpendicular (crossed) or parallel to the axis of the first polarizer.

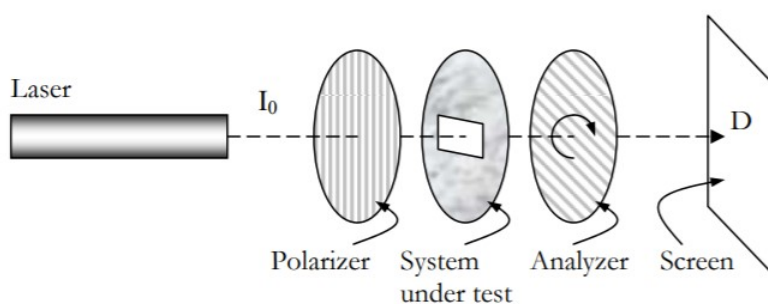


Figure 2. – Diagnostic two-polarizer system

1. With the first and second polarizers crossed in the above system, place a third sheet of linear polarizer in the system between the two stationary crossed sheets and observe the intensity of the transmitted light on the screen as you rotate the third polarizer. Make sure to record the orientation of every element in the diagnostic system in your lab notebook. Now place a photodetector at D and record below the intensity on the screen as a function of the angle of rotation of the third polarizer. About 10 points between (and including) two adjacent intensity maxima should be sufficient. Record this chart into your lab notebook.

Angle	Relative Intensity

2. Explain why the intensity on the screen changes as the central polarizer is rotated?

3. Derive the form of the theoretically expected intensity at the detector, and compare it with your measurement (you may use Jones matrices for this, or use basic trigonometry). Explain the discrepancy, if any.

1.3 Quarter and Half Wave Plates

1. You are given one quarter-wave plate and one half-wave plate (labeled A and B) but it is not known which plate is which. Use the system in Fig. 1 to: (i) find the axes of the quarter-wave plate, (ii) find the axes of the half-wave plate, and (iii) distinguish one plate from the other. Convince the T.A. that you have indeed found the axes and distinguished the two wave plates. Describe the experiments you performed on the two wave plates. Tabulate the results, and explain how you arrived at your conclusions. Make sure to record the orientation of every element in the diagnostic system in your lab notebooks.
2. A half-wave plate can be used to convert vertically polarized light into horizontally polarized light, and vice versa. Another useful application of a half-wave plate is as a linear polarization rotator. Indeed, if linear polarized light is incident at normal incidence on a half-wave plate and the plate is rotated through an angle $\mathbf{q}$ about an axis parallel with the propagation direction, the polarization of the transmitted beam is rotate by $2\mathbf{q}$. In your lab notebook, draw diagrams and explain how (1): a half-wave plate converts vertically polarized light to horizontally polarized light, and (2) how and why a half-wave plate achieves polarization rotation. (We are not looking for detailed calculations, but instead a detailed written description using diagrams as visual aids.)

1.4 Magic Sheet

You are given a sheet of "magic" material, which transmits light when folded in one direction, but blocks light when folded in the other direction. Observe this for yourself. Using only linear polarizers, devise nondestructive experiments to determine the possible optical elements that constitute the "magic" material. Record your data in tabular form in your lab notebook, and explain with the help of diagrams why light is blocked or passed in each experiment you perform. Show clearly how you arrive at your conclusion about the magic material.

1.5 Dielectric Reflection and the Brewster Angle

The Brewster angle is defined as the angle of incidence for which the reflected light intensity from a dielectric boundary goes to zero for TM polarization. This effect can be demonstrated by the experiment shown schematically below.

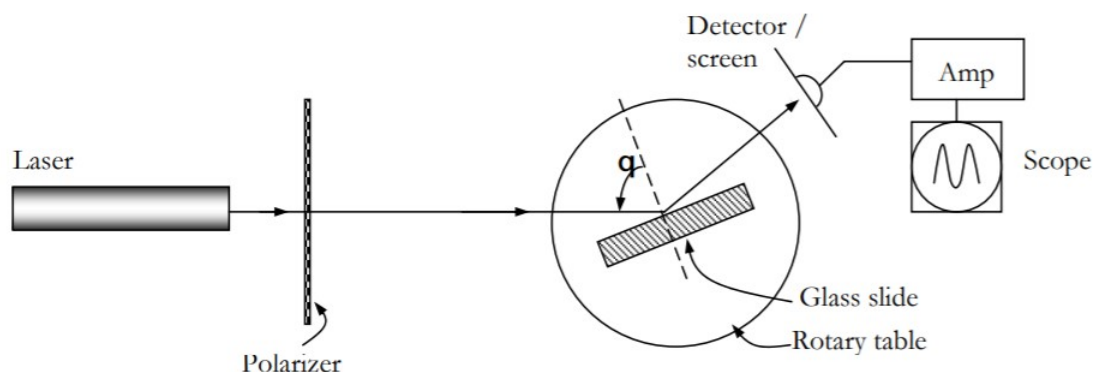


Figure 3. Experimental setup to demonstrate Brewster's angle.

Use a randomly polarized laser beam for this experiment. Align the glass slide so that its axis of rotation is perpendicular to the laser beam. This can be accomplished by adjusting the laser or the glass slide so that at normal incidence, the reflected beam goes back close to or into the laser cavity.

1. Where is the plane of incidence in your experiment?
2. With the linear polarizer aligned with its axis horizontal, and the laser beam centered on the axis of rotation of the glass slide, rotate the glass slide and observe the intensity of the reflected beam on the screen. Rotate the polarizer sheet by 90° and repeat the experiment. Copy the table below into your lab notebook. Record your observations of the reflected beam for each of the

two polarization cases in your lab notebook.

	Horizontal Polarization		Vertical Polarization	
Polarization in plane of incidence?	☐ Yes	☐ No	☐ Yes	☐ No
Does its intensity ever vanish?	☐ Yes	☐ No	☐ Yes	☐ No
If yes, corresponding angle of incidence	_____		_____	

3. Compute the index of refraction of the glass slide from your measurement of the angle. The wavelength of the He-Ne laser is 633 nm. Comment on the accuracy of this method of measuring refractive index.
4. Now, use the photodetector to measure the reflectivity of the glass slide as a function of the angle of incidence for both polarizations. About every 5° is good enough (the most accurate measurements are obtained if measurements for both polarizations are made before the angle of incidence is changed). Record the raw data in your lab notebook in a similar fashion to the table shown below. Plot your data using a software package of your choice, and paste it into your lab notebook.

TM Polarization

Angle of Incidence	Relative Intensity

TE Polarization

Angle of Incidence	Relative Intensity

1.7 (Part 1) Imaging Properties of a Convex Lens

In this exercise, we will explore the imaging properties of a simple symmetric biconvex lens. The setup we will use is shown in Figure 1. It consists of a white light source a transmission object (slide), a biconvex lens, and a screen, all mounted along an optical rail.

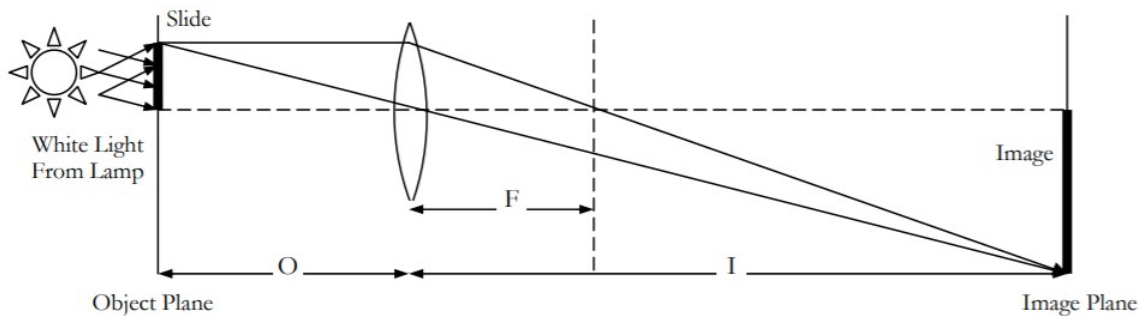


Figure 1. Setup to explore the imaging properties of a convex lens.

Using the setup in Fig. 1, and keeping the light source (slide) and object (lamp) fixed, select five different object distances, O , (distance from the lens to the object) by moving the lens. For each object distance, measure: (1) the corresponding image distance, I , for sharply defined image on the screen, and (2) the size of the image.

1. Make a plot of $1/I$ versus $1/O$ and from your plot determine the focal length of the lens (use all your data points – if you are not sure how to fit the data, please ask). Include the raw data, the plot and your calculation of the focal length in your report.
2. If the refractive index of the lens is 1.52, what is the radius of curvature of its convex surfaces?
3. Why is a sharp image not seen when the object distance is less than the focal length of the lens?
4. If two lenses of focal length F_1 and F_2 are abutted, what is the focal length of the combination? Show your calculation.
5. How would you go about measuring the focal length of a biconcave lens?

1.7 (Part 2) Optional for Extra Credit

Build the telescope you designed in the pre-lab exercise. Comment on the performance of your telescope. Did your telescope operate as you expected? Why or why not? Explain the operation of your telescope to the TA.

Congratulations, you finished the in-lab portion of Laboratory #1. Please write down in your notebook how long it took (start to finish) to complete: (a) the Pre-lab, (b) the in-lab portion, and (c) the postlab writeup.

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CHAPTER OVERVIEW

2: Coherence and Interferometry

[2.1: Pre-lab Exercises](#)

[2.2: In-lab Exercises](#)

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2.1: Pre-lab Exercises

Coherence and Interferometry

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PL 2.1 - Get Prepared to Start the Laboratory Exercises

Read the **entire** laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. **Write** them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade.

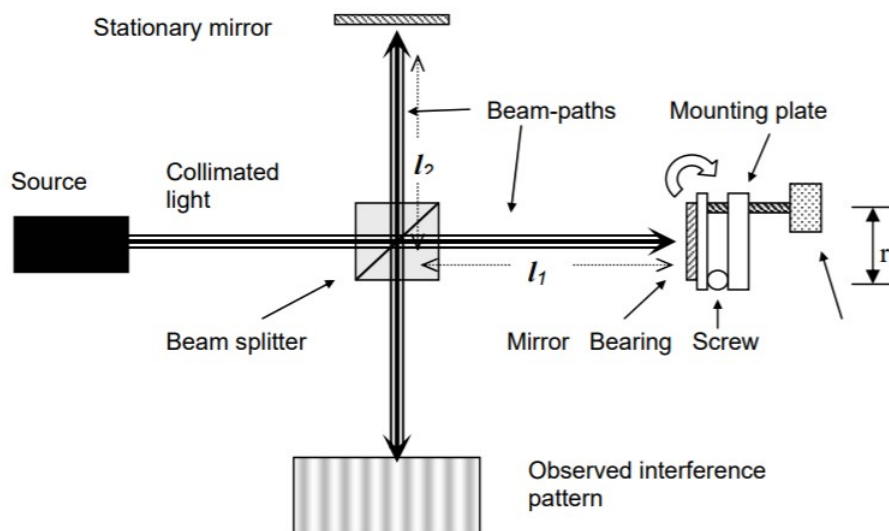
PL 2.2 - Michelson interferometer Considerations

For this exercise, you are greatly encouraged to work in groups of 2-4 people.

On Lab day, you will be asked to build and analyze one of the most commonly used interferometers in science and industry - a Michelson interferometer. Michelson interferometers are used in coherence detection apparatus, like FTIRs (Fourier Transform Infrared Spectroscopy), in countless mechanical systems (to maintain precise alignment), as well as in the proposed space-based Laser Interferometer Gravitational Wave Observatory (LIGO), just to cite a few examples. Michelson interferometers are relatively simple to build, easy to align and have many advantages over alternate interferometer designs. In this pre-lab exercise, you are asked to analyze a Michelson interferometer that will be used to measure the coherence length of a diode laser source. This interferometer has one stationary arm, and one translatable (moving arm) as shown in the diagram below.

First we would like you to determine the angular accuracy required to perform a 'rough alignment' of a Michelson interferometer.

1. Using collimated 632.8 nm (wavelength) light, you must determine the angular accuracy needed to form fringes of 1mm spacing on the test plane, given that the nominal arm length of the Michelson interferometer is $l_1 = l_2 = 20$ cm.
2. If the movable mirror is gimbaled from a ball bearing on one edge of a one-axis angular-rotation mount which uses a micrometer screw located at a distance $r = 2.5$ cm away at the opposite edge of the mirror, what is the corresponding distance the micrometer must travel to rotate the mirror between the angular alignment calculated above and the state of 'perfect' alignment (both output beams perfectly collinear)? Below is a diagram to illustrate the principle of the Michelson



interferometer.

3. If the light from the source is divergent (spherical waves instead of plane waves), what would the fringe pattern look like for a 'perfectly' aligned Michelson interferometer?

4. Although you are not being asked to do a full design of a coherence-detecting Michelson interferometer, you should think about how all of the parts listed in the table below could be used to build one. So please make a drawing in your lab notebook showing how all of the **optical** components listed below could be used to build a coherence-detection Michelson interferometer.

Table of Available Components

Qty	Description
2	1" two-axis mirrors
1	0.5" cube beam splitter
1	25 micron pinhole
2	Variable irises
1	Short focal-length 20x objective
1	Long focal-length biconvex lens (collimating lens)
1	3" linear translation stage w/micrometer
1 set	Posts, post holders and other mounting hardware
1	1'x1' optical breadboard
1	He-Ne laser used for alignment

"The strongest arguments prove nothing so long as the conclusions are not verified by experience. Experimental science is the queen of sciences and the goal of all speculation." -- Roger Bacon (1214-1294) English philosopher, scientist, optical experimentalist

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2.2: In-lab Exercises

Coherence and Interferometry

2.1 Mach-Zehnder Interferometer

You should find the Mach-Zehnder interferometer already set up and aligned. The instrument should be adjusted such that a uniform intensity pattern is produced on the screen (essentially no fringes). Make sure that the input He-Ne laser beam is well collimated by measuring the dimensions of the auxiliary output beam near the instrument and several meters away (the beam diameter should be roughly the same at both positions).

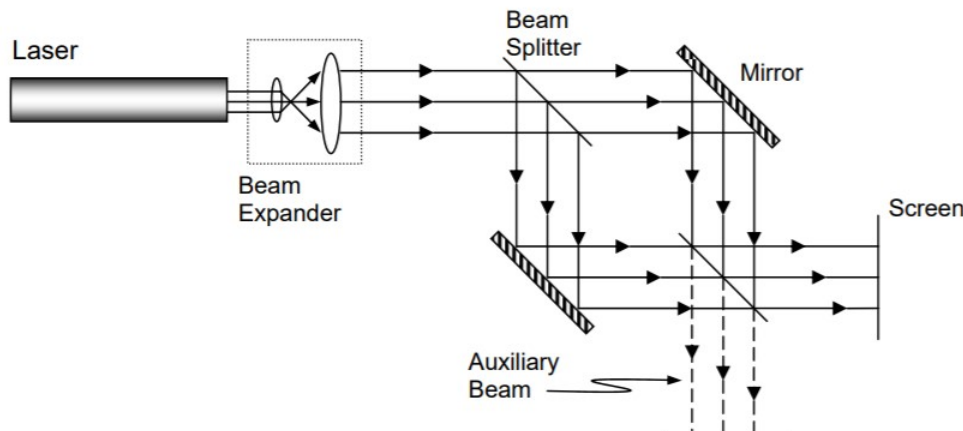


Figure 1. The Mach-Zehnder Interferometer

For this exercise, we will work with a glass slide (about 2mm thick) that has negligible curvature, and therefore we shall approximate the surfaces of the slide as planar, but non-parallel. The slide, therefore, has the shape of a wedge as illustrated in Figure 2 (b). Now, insert the given microscope slide into the mount, and place the mount in one of the arms of the interferometer.

1. In your notebook, make a sketch of the fringes on the screen, measure the spacing between the dark fringes and label your sketch with the distances (spacings) you measured.
2. Use your data to calculate the parallelism of the glass slide. That is, determine the angle θ between the surfaces of the wedge (the wavelength of the He-Ne laser light is 632.8 nm).
3. By now you should have observed that the two output beams of the Mach-Zehnder interferometer are complementary. Explain with the aid of the previous diagram and your knowledge of phase changes on reflection from surfaces why this is so.

2.2 Flatness Measurements

Normally, we think of a glass slide as a thin slab with boundaries defined by two perfectly parallel planes. Of course, in reality this is not the case. The glass-air boundaries may be curved, undulating, or planar, and in general are not parallel. Curvature may be in one or many directions as illustrated in Figs. 2 (a) and 2 (b).

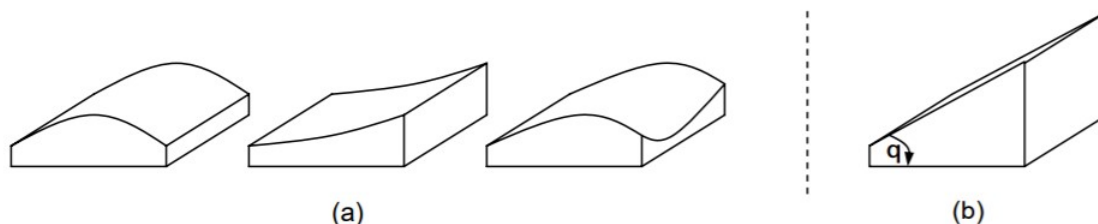


Figure 2. (a) Possible shapes of curved glass slides; (b) A wedged glass slide

The goal of this experiment is to qualitatively determine the surface flatness of a number of glass-slide samples. The equipment to be used includes a sodium vapor lamp (quasi-monochromatic source), a reference flat (glass block with a reference surface of known flatness), lens paper, and the samples to be studied. The setup is as illustrated in Figure 3.

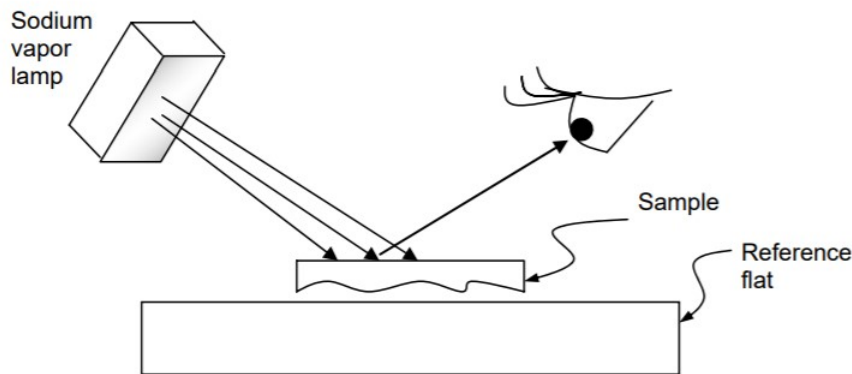


Figure 3. Setup for measuring the flatness of glass blocks and plates

Place the surface of the sample to be measured against the reference flat and illuminate the assembly with the sodium lamp. You should observe a distinct set of fringes, which illustrate the relative flatness of the surface of the sample, assuming the reference flat is perfectly flat. Dust and other foreign particles between the two surfaces will lead to a wedge of air between the sample and the flat and to a fine fringe pattern that will be difficult to interpret. If this occurs, place a sheet of clean lens paper between the two surfaces and slowly drag the lens paper out from between the surfaces. This should remove most of the loose dust particles. Rotate the sample slowly until the widest possible spacing between the fringes is obtained.

For (a) the plexiglass triangular block (b) the rectangular glass block, (c) the wedged glass block, and (d) the thick circular glass disc:

1. Sketch and describe in your notebook the fringes you get for each of the four samples.
2. Interpret the fringes in the context of the flatness of the surfaces by drawing pictures in your notebook of the surface profiles, and
3. Rank the four samples in increasing order of flatness.

2.3 Parallelism Measurements

Several numbered thin cover glass slides (about a dozen) are provided. A simple qualitative method of quickly establishing parallelism of the glass slides involves the examination of Haidinger fringes as described next. Place several of the thin glass slides on the clean sheet of paper, which lies on the small low table. Stand upright over the glass slides with the sodium light next to your forehead and illuminate the glass slides so that interference patterns are observed for each glass side. By comparing the interference patterns (Haidinger fringes) identify: (a) the most plane parallel, and (b) the least plane parallel glass slide. Write the numbers of the two slides in your notebook. Explain to the TA how you arrived at your choice and have the TA sign your notebook.

1. Explain how Haidinger fringes are formed.
2. What should the Haidinger fringe pattern look like for a perfectly parallel glass slide?

2.4 Multiple Beam Interferometry with a Thin Glass Plate

In this experiment, we will attempt to observe the gross features of multiple-beam interference as described theoretically in the notes. The setup to be used is shown in Fig. 4. It consists of a thin glass plate (Corning #1 cover glass slide) that is mounted on a rotary table such that it can be illuminated with collimated He-Ne laser light ($\lambda = 0.6328 \mu\text{m}$) at varying angles of incidence (the setup uses an unexpanded He-Ne laser beam that is about 2 mm in diameter). Two screens are used to view the reflected and transmitted interference patterns. The thickness of the plate is about 150 mm.

We will vary the external angle of incidence, $\mathbf{q_i}$, by rotating the glass slide using the rotary table, and observe the interference patterns on the screens. Note that the range of $\mathbf{q_t}$ (the internal angle) is limited from 0 to $\mathbf{q_c}$, the critical angle. Unfortunately, varying $\mathbf{q_i}$ causes both d , the path difference between successive emerging beams, and R , the surface reflectivity, to change at the

same time. However, in this experiment it is still possible to observe the effect of varying R on the sharpness (separation/width) of the fringes. Varying d simply brings fringes of different order into view; these may be spaced closer or farther apart on the screen depending on q_i .

Specifically, you should observe that for small values of q_i , R is small, and so the finesse, F , is low -- hence, one cannot tell whether the pattern consists of bright fringes on dark background or vice-versa. However, as q_i increases, R increases and, therefore, the finesse and the fringe contrast increases. Since the general tendency must be an increase in reflectance (reduction in transmittance) as q_i is increased, the reflected beam will, therefore, contain most of the energy when the finesse is high except at such angles where this condition is violated by destructive interference. Hence the reflected beam will consist of dark fringes on a bright background.

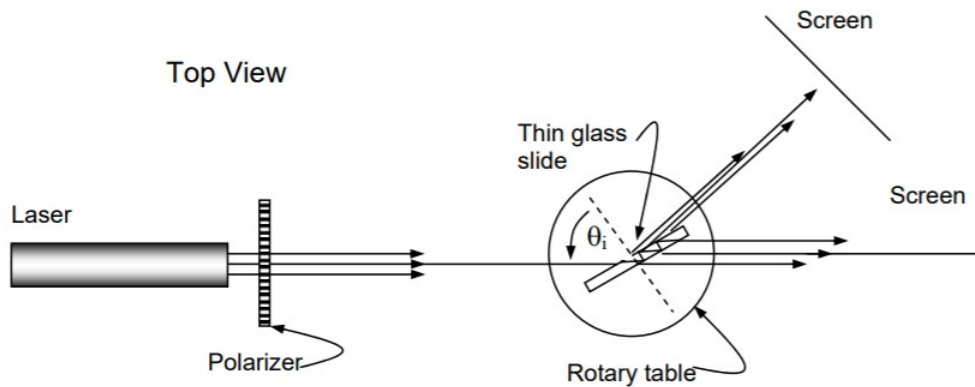


Figure 4. Setup for observing multiple beam interference from a thin glass slide

1. With the laser beam incident on the region near the center of the plate, rotate the plate, and with the polarizer axis vertical (TE wave), rotate the glass plate and observe the transmitted and reflected beams on the screens. In particular, note the angular separation of the fringes in the reflected beam for each of the following excitation conditions: (i) just above normal incidence, (ii) in the neighborhood of 45° angle of incidence, and (iii) near grazing incidence. Which of the three positions gives the smallest fringe separation? Why? [Hint: consider dd/dq].
2. Explain why near grazing incidence the reflected interference pattern consists of dark fringes on a bright background, while near normal incidence the pattern could be interpreted as either dark fringes on a bright background or vice versa.
3. What important statements can you make concerning the complementarities of the reflected and transmitted beams? Justify your statements.

2.5 Fabry-Perot Interferometer

The setup for the Fabry-Perot interferometer is shown in Figure 5. You will find the instrument already set up and aligned when you enter the Lab. If a well-defined circular fringe pattern is not observed on the screen, remove the diffuser and adjust one of the mirror micrometers so that the beam does not "walk off" to one side of the instrument, and then replace the diffuser. A circular fringe pattern should now be observed on the screen. Place the screen in the focal plane of the lens, and observe the fringe pattern.

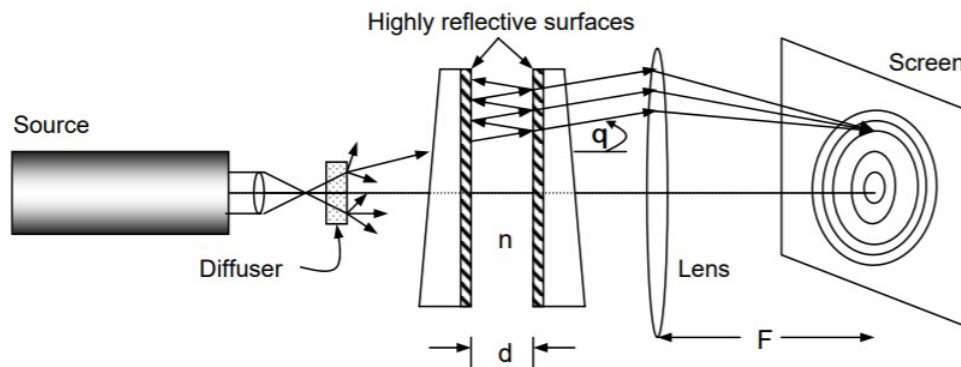


Figure 5. The Fabry-Perot Interferometer

1. Tape a sheet of paper (your personal screen) over the screen, trace out the first few innermost fringes, and paste your personal screen into your notebook. Measure the distance F .
2. Record your data, from part (a), for the measured fringe positions, calculated angular positions, and their orders ($\dots m-1, m, m+1 \dots$) in tabular form in your notebook. Label your sketch with the measured angles.
3. Write the condition for a bright maximum.
4. Explain why circular fringes are obtained.
5. From the data in (a) determine the order of interference of the innermost bright ring and the thickness d of the air gap. Hint: *Since each circular fringe is characterized by its own m and q , by taking measurements on two fringes (preferably not adjacent) and noting that for the pair you can write*

$$m_1\lambda = 2nd \cos q_1 \text{ and } m_2\lambda = (m_1 - Dm)\lambda = 2nd \cos q_2$$

where Dm is the difference in order between the two fringes, it then becomes possible to solve for d and m_1 .

2.6 Lummer-Gehrcke Interferometer

The Lummer-Gehrcke interferometer is illustrated in Figure 6. In our case, it is a thin glass plate with a light-coupling prism cemented onto one end of the plate. You will find the interferometer sitting on a heavy metal block, and the laser on a lab jack. Under no circumstances should you touch the interferometer. The plate is very fragile and, furthermore, fingerprints seriously degrade its performance.

In order to accurately measure the angular positions of the fringes, the reference point b on the screen (where the top plane of the interferometer intersects the screen) must be found. This point is found by adjusting the height and angle of the laser beam so it just grazes the top surface of the interferometer along the entire length of the plate, and identifying the point b where the laser beam strikes the screen. Tape a sheet of paper (your personal screen) over the screen and mark the point b . The accuracy of your final results will depend critically on how well you were able to find the point b .

With the laser beam still grazing the plate, place the lens at the end of the plate as shown in the diagram, and adjust the position of the lens so that the beam hits the same point b on the screen. This assures that the axis of the lens lies on the top surface of the plate. The screen should also be positioned one focal length behind the lens.

Now lower the lab jack back to its normal position so that the laser light is coupled into the plate via the prism. Adjust the laser beam so that the light reflected from the back of the prism returns close its original path back to the laser.

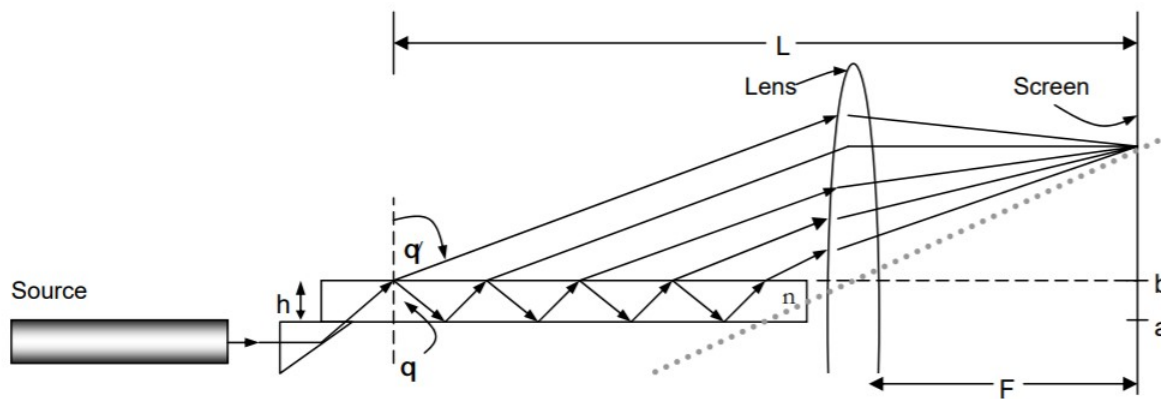


Figure 6. The Lummer-Gehrcke Interferometer

1. On your personal screen mark the location of at least the first six fringes above the point b. Measure the distance F (the focal length of the lens). This should also be the position where the fringes are sharpest. In your lab notebook make a table of the fringe positions, their calculated angular positions, their orders ($\dots, m-1, m, m+1, \dots$) for the first 6 fringes.
2. Assuming that with the lens each fringe is formed by interference between successive emerging beams with the same value of q , write the interference condition for a bright fringe.
3. From your data determine: (a) the order of interference, for the first bright fringe above the point b, and (b) the index of refraction of the glass at 632.8nm. Assume that the thickness of the plate is 150 mm.
4. Remove the lens and re-record the position of the fringes. Explain the role of the lens in the previous setup, and the observed differences between the fringe pattern with the lens in place and the pattern without the lens. Under what conditions would the two patterns yield the same angular data for the position of the fringes?
5. Although the He-Ne laser is fairly well collimated, the interference pattern in our LummerGehrcke interferometer is visible over a considerable range of exit angles. Explain why.
6. If a thin, isotropic film of refractive index n_2 and thickness t_2 , where t_2 is much larger than the penetration depth, was deposited on the plate, what would be the new condition for interference? Draw a new ray optics picture to help you see what is going on. How would you expect the fringe pattern to change?
7. The plate thickness of your interferometer is about 33 times less than that of the original instrument of the 1920's. Using the results of (f), which interferometer has the higher resolution $[(Dq_m)/Dt_2]$ when used for measuring thickness and index of refraction of thin films placed on the plate? (See C. Warde, Applied Optics 15, 2730 (1976))

2.7 Michelson Interferometer (Optional)

Build the interferometer you drew in the pre-lab exercise. Comment on the performance of your interferometer. Did your interferometer operate as you expected? What were some of the difficulties you encountered while aligning it? Explain the operation of the coherence-length detection interferometer to the TA.

If time permits, use your system to record coherence-length data for a diode laser (It is at the TA's discretion whether or not you take this measurement).

*"I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind." -- Lord Kelvin (William Thomson, 1st Baron) (1824-1907) English physicist and mathematician. In: **Popular Lectures and Addresses**, London, 1889, v. I, p. 73. See also: **Life of Lord Kelvin**, by S. P. Thompson, 1910, V. 2, p. 792.*

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CHAPTER OVERVIEW

3: Diffraction

[3.1: Pre-lab Exercises](#)

[3.2: In-lab Exercises](#)

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3.1: Pre-lab Exercises

Diffraction

The Pre-Lab Exercises must be completed BEFORE entering the Lab. In your lab notebook record data, explain phenomena you observe, and answer the questions asked. Remember to answer all questions in your lab notebook in a neat and orderly fashion. No data are to be taken on these laboratory sheets. Tables provided herein are simply examples of how to record data into your laboratory notebooks. Expect the in-lab portion of this exercise to take about 3 hours.

PL 3.1 – Get Prepared to Start the Laboratory Exercises

Read the entire laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. **Write** them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade.

"If you cannot saw with a file or file with a saw, then you will be no good as an experimentalist." – Augustin Fresnel (1788-1827)

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3.2: In-lab Exercises

Diffraction

Introduction

The objectives of these laboratory exercises are to explore and investigate the phenomenon of diffraction through a series of simple and then more advanced experiments. The first set of diffraction experiments will be carried out with He-Ne laser light with a wavelength of 632.8 nm. These include:

1. Fraunhofer diffraction from single slits, double slits, multiple slits, gratings, circular, rectangular, triangular and hexagonal apertures, and rectangular grids.
2. Fresnel diffraction from circular apertures and obstacles.
3. Investigation of the focusing and imaging properties of a Fresnel zone plate.

A. Fraunhofer Diffraction

The setup we will use to demonstrate and study Fraunhofer diffraction is shown below. Note that it is important to illuminate the objects with a planar wavefront to observe Fraunhofer diffraction. In many cases, a reasonably accurate Fraunhofer pattern can be obtained with the raw laser beam (i.e., without the beam expander) and with the screen sufficiently far away from the diffracting object (see Figure 1).

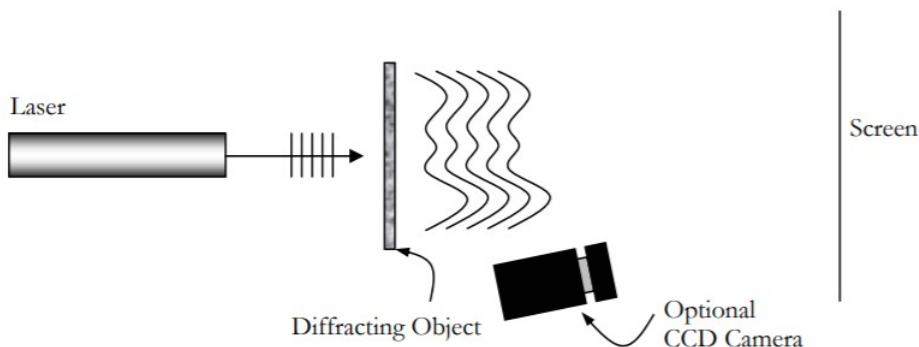


Figure 1. Setup for observing Fraunhofer Diffraction

3.1 Single Slit

1. Set up and observe the Fraunhofer diffraction pattern owing to the variable single slit. Observe how the pattern changes as the slit-width is varied. Qualitatively sketch the typical pattern. Describe your observations. Explain why the pattern changes the way it does.
2. What is the theoretical expression for the angular location of the bright fringes?
3. Have the TA set the slit width to a specific value. Record the corresponding micrometer reading in your notebook. Measure the angular distribution of the bright fringes. Record the measurements in your notebook and determine the slit width from your measurements.

3.2 Double Slit

Set up and observe the Fraunhofer diffraction pattern owing to the given double slits. These are contained on the upper level of the plastic slide labeled 3.2/3.3.

1. Sketch the pattern, and measure the angular distribution of the bright fringes for the double slit with the widest separation. From your measurement calculate the slit width and the slit separation
2. In addition, view the other double slits of varying separation on the top level of the same plastic slide. Qualitatively sketch your observations. Rank the double slits in order of increasing separation.
3. Explain why and how varying the slit separation affects the diffraction pattern.
4. *Optional:* Using the CCD imager, as shown in Figure 1, capture an image of the two-slit diffraction pattern. It is important that you not move the imager from its current location. Using the captured image, employ Matlab to calculate the relative spacing of

the slits. Provide a reconstruction of the slits, then compare their relative dimensions numerically.

3.3 Multiple Slits

Set up and observe the Fraunhofer diffraction pattern due to the N-slit gratings. These are contained on the bottom row of the plastic-mounted slide of 3.2. The number of slits, N, in each grating is indicated beneath each set of the gratings.

1. Sketch the pattern for various values of N and explain **qualitatively** what happens as N increases from 2. What is the theoretically expected pattern as $N \rightarrow \infty$.
2. Examine the Fraunhofer patterns from the three amplitude transmission diffraction gratings (labeled as 2,400, 7,500 and 15,000 lines/inch) and describe the patterns you see. Do the patterns from the gratings behave as predicted by the theory?
3. For the grating labeled 15,000 lines per inch, measure the diffraction angles of the spots, and use this information to compute the actual periodicity of the grating. The grating material has aged over the past 10 years because of environmental effects. Did it shrink or expand?
4. Note that only 5 diffracted spots are visible with the 15,000 lpi grating. Why do we not see an infinite number of spots? What determines the maximum number of diffraction spots? Derive a mathematical expression for the maximum number of spots.

3.4 Rectangular Grid

Set up and observe the Fraunhofer diffraction due to the rectangular grid. Make a qualitative sketch of the pattern and explain how it differs from that of the rectangular aperture (below).

3.5 Rectangular Aperture

Set up and observe the Fraunhofer diffraction pattern due to the rectangular apertures. There are five different rectangles on the plate. Refer to the diagram below to find their location on the plate. Illuminating the plate from behind with an incoherent light source (e.g., a flashlight) should prove helpful in locating and aligning the apertures in the plate.

1. Choose any rectangular aperture and sketch its diffraction pattern on a separate sheet of paper (to be pasted into your lab notebook). Make angular fringe location measurements and label your sketch with the measurements.
2. Use your measurements to compute the width (horizontal) and the height (vertical) of one of the apertures. If needed, have the TA or LA assist you in finding the diffraction objects on each plate provided. Size shown below is exaggerated.

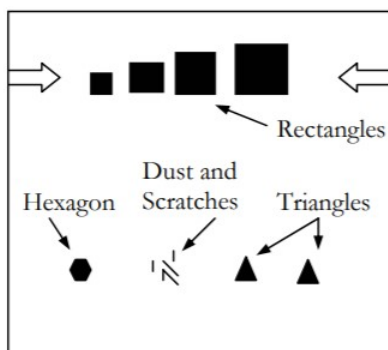


Figure 2. Glass Plate for 3.5 and 3.6

3.6 Triangular and Hexagonal Apertures

Set up and observe the Fraunhofer diffraction pattern owing to the triangular and hexagonal apertures. These are located on the second row of the plate as illustrated in Figure 2.

1. Roughly sketch the patterns next to each other in your lab notebook.
2. What are the qualitative differences between the patterns?
3. Using the Matlab simulation of the hexagon and triangle diffraction intensity patterns (in Problem set 3), describe the difference between the two simulated diffraction patterns. What are the observable differences between the computer simulation, and the actual observed pattern?

3.7 Circular Aperture

Set up and observe the Fraunhofer diffraction pattern due to the circular aperture.

1. Qualitatively sketch the pattern, but quantitatively measure the diameter of the Airy disc, and label the sketch with your measurement. Make sure you measure the Airy disc correctly! Check with the T.A. or L.A. to be sure. In your lab writeup, be sure to include the sketched pattern, and indicate precisely the boundaries of the Airy disc.
2. Use the results of (a) to calculate the size of the aperture.

3.8 2-D Grating Mirror

The thin substrate consists of a 2-D array of circular holes in a mirror substrate. Operate the system in reflection with an angle of incidence of 45° . Set up and observe the Fraunhofer pattern due to this grating.

1. Sketch the diffraction pattern.
2. What is the packing symmetry of the holes in the mirror?
3. Measure the dimensions of the pattern, and label the sketch with the dimensions.
4. Compute the pitch of the grating.
5. Is this structure a phase grating, or an amplitude grating? Why?

B. Fresnel Diffraction

The setup for Fresnel diffraction is shown below. The wavefront incident on the apertures must be very nonplanar and the screen-to-object distance small (5 to 40 cm) to obtain dramatic results.

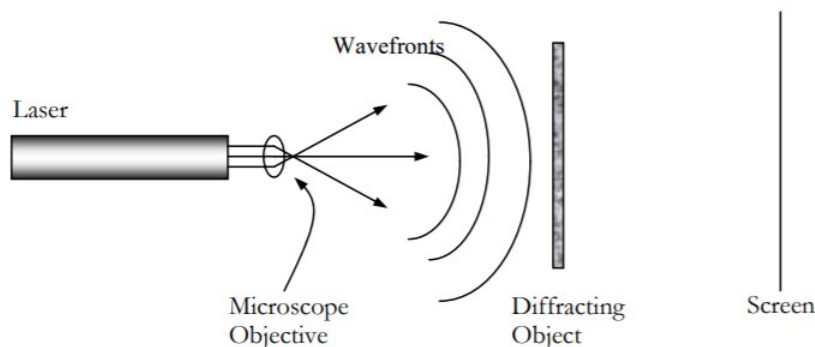


Figure 3. Setup for observing Fresnel Diffraction

3.9 Circular Opening

Set up and observe the Fresnel diffraction pattern due to the small circular opening. Vary the distance between the source and the aperture. The circular aperture must be less than 1 cm away from the objective lens to see the Fresnel diffraction pattern.

1. Sketch a few of the different patterns you get.
2. Compare the Fresnel diffraction patterns with the Fraunhofer pattern from the circular aperture.
3. Explain how and why the Fresnel patterns are different from the Fraunhofer patterns.

3.10 Circular Obstacle

Set up and observe the Fresnel diffraction patterns from the small spherical obstacle.

1. Qualitatively sketch the pattern.
2. Explain why there is always a bright spot in the center of the dark shadow.

3.11 Fresnel Zone Plate: Measurement of focal lengths

1. Place the Fresnel zone plate in a well-collimated beam of He-Ne laser light and measure the positions of the farthest three or four focused spots you can see.
2. From the data in (a) what are the primary, secondary and tertiary focal lengths (F_1 , F_2 , and F_3 , respectively) of the zone plate? What are your measured ratios: F_2/F_1 , F_3/F_1 and F_3/F_2 ? Are these in agreement with the theoretical values? If not, why?

3. From your data in (b) calculate the radius of the first zone of the zone plate. Measure the diameter of the first zone to the nearest half-millimeter with a ruler. Explain any significant difference between your measured and calculated values.

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CHAPTER OVERVIEW

4: Holography

[4.1: Pre-lab Exercises](#)

[4.2: In-lab Exercises](#)

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4.1: Pre-lab Exercises

Holography: An Example of Diffraction

The Pre-Lab Exercises must be completed BEFORE entering the Lab. In your lab notebook record data, explain phenomena you observe, and answer the questions asked. Remember to answer all questions in your lab notebook in a neat and orderly fashion. No data are to be taken on these laboratory sheets. Tables provided herein are simply examples of how to record data into your laboratory notebooks. Expect the in-lab portion of this exercise to take about 3 hours.

PL 4.1 – Get Prepared to Start the Laboratory Exercises

Read the **entire** laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. **Write** them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade.

PL 4.2 Preparation for Making Holograms

In Lab Exercise 4.2, your group will be making one type of hologram. You will have the option of making: (a) a Denisjuk single-beam white-light reflection hologram (simplest), (b) a two-beam or multiple beam reflection hologram, (c) a two-beam transmission hologram (must be viewed with a laser) or (d) a rainbow hologram (pseudo-color – but it takes quite a while to setup and develop). For those with little or no experience with optics or holography, this is a great opportunity to get your feet wet (pun intended).

For this section of the Pre-Lab it is your job to find out more about these types of holograms - everything about exposure of the holographic film, its development, and processing will be explained on lab day. However, before coming to the Lab you are advised to do some outside research on how to configure the holographic recording and readout setups for each of the four types of holograms listed above.

After you have decided which hologram you want to make, make a detailed sketch of the recording system you will use. – this includes placement of the object, lenses, laser, mirrors, beam splitters, holographic plate, and any other items you might need (e.g., slits, wave plates, irises, etc...).

Here are some useful tips to consider before coming to the MOL to make holograms:

1. You should bring some *interesting* objects from which we can make holograms. Such objects should not be dark in color, nor should they exceed a volume of 3"x3"x3".
2. Those students whose objects we use, will be able to take home a copy of their hologram. Otherwise, holograms made in the MOL will be archived for demonstration to future students. You will need your own laser to view the transmission holograms at home. Reflection holograms will be viewable in white light.
3. Make sure to wear clothes that can handle stains. Also, you will not be able to leave to go to the bathroom while the plates are developing, so be sure to go beforehand.
4. Be aware that it will be extremely dark at times in the laboratory, and that it can be fairly timeconsuming depending on the complexity of the hologram you choose to make.

"By-the bye, what became of the baby?" said the Cat. 'I'd nearly forgotten to ask.' 'It turned into a pig,' Alice answered." – Lewis Carroll, Alice in Wonderland

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4.2: In-lab Exercises

Holography: An Example of Diffraction

4.1 Viewing of Transmission, Reflection, White-Light and Computer-Generated Holograms

A collimated laser beam and a white-light source have been set up for viewing of four types of holograms.

1. Illuminate the given laser-made **transmission holograms** with the divergent laser beam (remove the large collimating lens). For each hologram, find the direction of propagation of the virtual image, and view it directly (i.e., no lens or screen is to be used to view this image). Move your head around and tilt the hologram to get the best view of the image. For the **hologram of the clock or the hologram of the lion**, draw a diagram of the setup that gives a virtual image with the best fidelity. In your diagram, show the location of the virtual image and any other beams that exit the hologram. Use this information to infer the geometry of the setup that was used to record the hologram (reverse engineering).
2. Illuminate the **reflection hologram of the coins** with the divergent laser beam and view the image through the hologram in reflection. Is the image you see real or virtual? Draw a diagram of the setup showing the location of the image and any other beams that exit the hologram. Use this information to infer the geometry of the setup used to record the hologram.
3. Illuminate the **white-light hologram of the owl** with white-light from the overhead lights in the Laboratory such that the light strikes the hologram and then reflects an image into your eye. Tip the hologram from side to side and up and down. Draw a diagram of your readout configuration showing the location of the image or images. Describe your observations from tipping the hologram in the beam. Re-illuminate the white light hologram with the divergent He-Ne laser beam. What differences do you observe? Explain these differences. Use your observations to infer the geometry of the setup used to record this hologram.
4. Put the collimating lens back into the system so as to generate a collimated laser beam. Illuminate the **first computer-generated transmission hologram** (written on a 4"x5" plate) with the collimated laser beam, and view the real image (it will spell a familiar acronym) on a distant screen. Draw a diagram of the setup showing the location of the real image and any other beams that exit the hologram. Use this information to infer the geometry of the setup used to record the hologram.
5. Place the **second computer-generated hologram** (written on 2"x2" film) into the collimated laser beam. Now relocate the screen close to second computer-generated hologram. Move the screen away from the hologram slowly. What do you observe on the screen? How do you explain what you see on the screen with respect to what you see on the actual hologram? Ask the TA or LA to show you how this hologram was made.

4.2 Making Your Hologram

In Prelab 4.2, you studied the recording setup for the following four types of holograms: (a) Denisyuk single-beam white-light reflection hologram, (b) two-beam or multiple beam reflection hologram, (c) two-beam transmission hologram and (d) rainbow hologram. Working with your group and with the assistance of the TA, now make one or more copies of the hologram type you chose.

1. In your lab notebook, describe and draw a diagram of the setup you used to make your hologram. Measure all distances between the optical and mechanical components in your setup, and show these dimensions in your diagram.
2. If possible, also specify: (a) the laser power density in the object and reference beams at the recording medium, and (b) the type, model number and manufacturer of the recording medium as appropriate.
3. Comment on the quality of the hologram that you made, and explain what could or should have been done to improve its quality.
4. Was this experience worthwhile?

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CHAPTER OVERVIEW

5: Electro-optic, Acousto-optic and Photorefractive Light Modulation

[5.1: Pre-lab Exercises](#)

[5.2: In-lab Exercises](#)

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5.1: Pre-lab Exercises

Electro-optic, Acousto-optic and Photorefractive Light Modulation

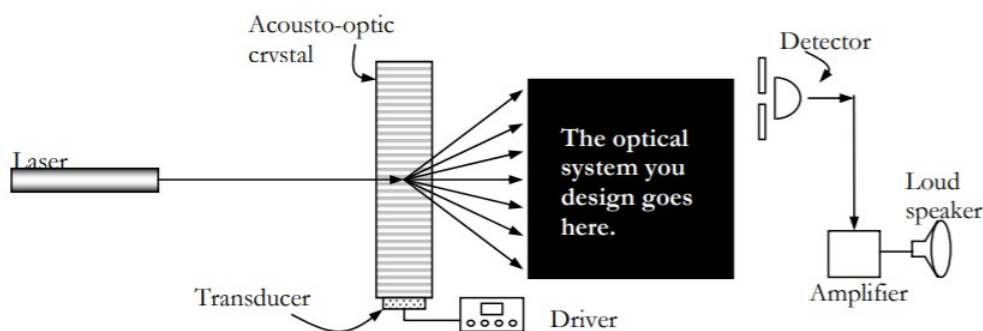
The Pre-Lab Exercises must be completed BEFORE entering the Lab. In your lab notebook record data, explain phenomena you observe, and answer the questions asked. Remember to answer all questions in your lab notebook in a neat and orderly fashion. No data are to be taken on these laboratory sheets. Tables provided herein are simply examples of how to record data into your laboratory notebooks. Expect the in-lab portion of this exercise to take about 4 hours. **You may do Exercises 5.1 – 5.3 this week and Exercise 5.4 next week.**

PL5.1 - Get Prepared to Start the Laboratory Exercises

Read the entire laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. **Write** them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade. **In the electro-optic light modulation exercise we will be setting up an optical (wireless) communications system (see Exercise 5.26). If you have a CD-player and a CD you would like to use as an input source of music for transmission over the laser beam, bring it with you on Lab day.**

PL5.2 (optional)

In exercise 5.24, you are asked to perform an optical communication system demonstration. For this exercise, you will be using a beam modulated in spatial frequency (the light will sweep back and forth across a distant screen at the same frequency as an audio signal input). However, using this form of spatial light modulation is not conducive to high-fidelity audio (since a detector is very small in comparison to the size of the beam swath – you will hear a series of clicks instead of a clear, smooth audio waveform from the detector/speaker assembly). Using optical components that you are already familiar with, describe (mathematically if necessary) how to convert the spatial frequency modulation into an amplitude-only modulation. Design the system that replaces the black box below.



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5.2: In-lab Exercises

Electro-optic, Acousto-optic and Photorefractive Light Modulation

5.1 Propagation of Light through Anisotropic Crystals

5.1.1 Calcite crystal

1. The Calcite crystal has a trapezoidal shape. Place the calcite crystal on a page of printed text and observe the doubly refracted light as evidenced by the existence of two images of the text.
2. Place the calcite crystal at an oblique angle between the crossed-polarizer system as shown in Fig.1
3. Rotate both the input and output polarizers to see if you can find a joint polarizer orientation that extinguishes one of the two transmitted beams. After you have found this orientation, keep the output polarizer fixed. Now rotate the input polarizer and observe the changes in the intensity of the two transmitted beams. Record your observations in your Lab notebook.
4. From your data, determine which beam is the extraordinary ray.
5. Is this crystal positive or negative uniaxial? Explain your answer.

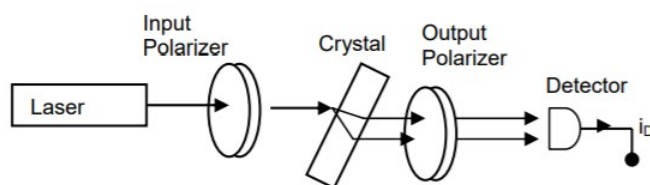


Fig. 1. Setup to observe ordinary and extraordinary ray propagation in a calcite crystal

5.1.2 Finding the Optical-Axis of a Barium Titanate Crystal

You are provided a barium titanate crystal (BaTiO_3) that has a roughly square cross-section. You are also provided 2 sheets of polarizers, and a randomly-polarized light source (an incandescent light bulb).

Barium titanate is an optically anisotropic ferroelectric crystal. The crystal is uniaxial, and so it is characterized by two indices of refraction. The axes of the crystal are labeled, a, b, and c, where the c axis is the axis of highest symmetry (four fold in this case). The c axis is also the ferroelectric axis.

The c axis can be determined by observing the transmission characteristics of the crystal in polarized white light. The setup, which is illustrated in below, consists of a white-light source and a pair of crossed linear polarizers.

The four-fold c-axis is optically isotropic which means that light propagating along the c axis will remain extinguished as the crystal is rotated between the crossed polarizers. In this case, the light "sees" the ordinary refractive index, n_o , for all orientations of the polarization vector of the light within the a-b plane.

On the other hand, the a and b axes each have two-fold symmetry and light propagating along the a axis will "see" n_o when it is polarized along b, and n_e , the extraordinary refractive index, when it is polarized along c. With the crystal between crossed polarizers and the light propagating along the a or b axis, extinction should occur every time the crystal is rotated by 90° about the propagation direction.

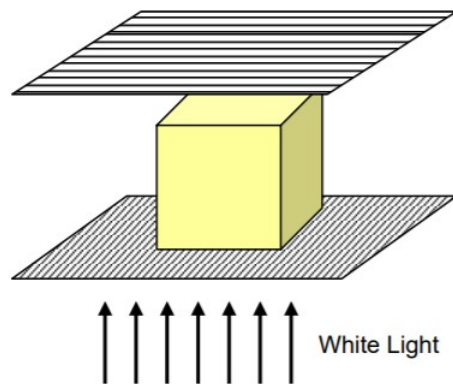


Fig. 2. Crossed polarizer setup for finding the c -axis of the crystal.

1. Find the c axis of the barium titanate crystal by carrying out the procedures described above. Convince the TA or LA that you have found the c -axis.
2. Why does light with its k vector parallel to the c -axis of the Barium Titanate crystal remain extinguished as the crystal is rotated about the c -axis between crossed polarizers?
3. Why is light with its k vector parallel to the a -axis of the crystal extinguished every time the barium titanate crystal is rotated between crossed polarizers by 90° about the a -axis? In your answer, be sure to explain what happens at intermediate angles of rotation.
4. From your answers in (1), (2) and (3), what are the possible crystal symmetries (e.g., cubic, monoclinic, etc.) that would give the same result?
5. What are the actual crystal symmetries of Calcite and BaTiO_3 ? What are their permittivity (**NOT electro-optic**) tensors (this may require a little bit of outside research)?

5.2. Electro-optic Modulation

A lithium niobate (LiNbO_3) crystal will be used to modulate the intensity of a He-Ne laser beam. The setup is shown in Figure 3.

CAUTION: THE HIGH VOLTAGES APPLIED TO THE MODULATOR CAN BE LETHAL; ALWAYS TURN OFF THE HIGH-VOLTAGE POWER SUPPLY BEFORE TOUCHING MODULATOR OR ATTEMPTING TO DISCONNECT THE WIRES.

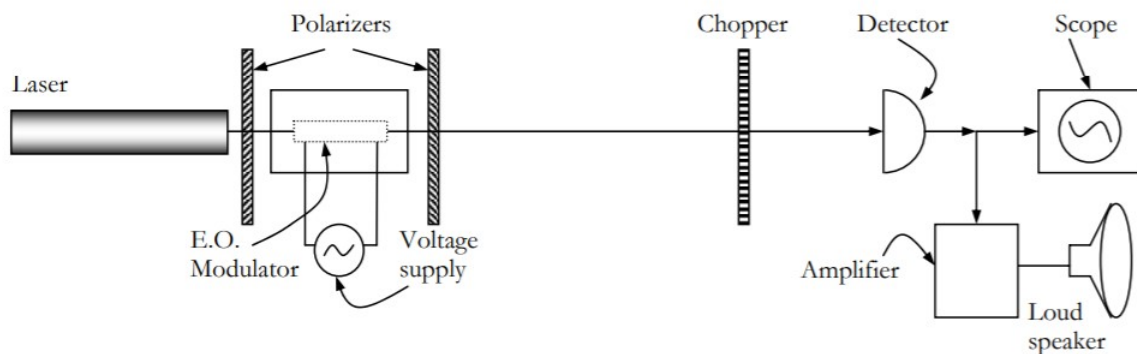


Figure 3. Setup for studying electro-optic modulation of light in a LiNbO_3 crystal

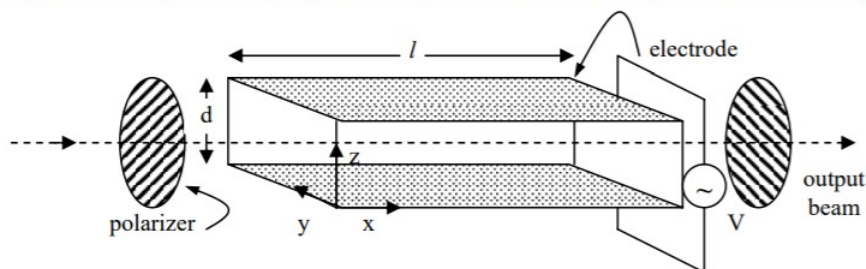


Figure 4. Crystal cut and readout configuration for the LiNbO_3 modulator.

5.2.1. Please answer the following questions relating the transverse LiNbO_3 Modulator

1. What are the approximate numerical values of the refractive indices of the LiNbO_3 crystal in the region of the spectrum close to the He-Ne laser wavelength of 633 nm? (You can find this information in Yariv)
2. Is it a negative uniaxial or a positive uniaxial crystal?
3. Write down the form of the electro-optic tensor for the LiNbO_3 crystal clearly showing which tensor elements are zero.
4. The crystal cut and readout configuration for your modulator is as shown in Fig. 4. Derive the expression for the phase retardation $G = f_y - f_z$.
5. The LiNbO_3 crystal in your setup is about 2 cm long, what is the calculated half-wave voltage?

5.2.2 Modulator Principal Axes

Before applying voltage to the modulator, make sure the modulator is placed between a pair of crossed polarizers as shown in Fig. 1, and use an unpolarized He-Ne laser beam in the setup to read out the modulator. Keeping the polarizers crossed, slowly rotate them together about the propagation direction in increments of about 5° and observe the intensity of the light transmitted through the second polarizer (analyzer). From these observations, determine the principal axes of the modulator crystal. Convince the TA or LA that you have found the axes.

5.2.3 Modulator Transmission Versus Applied Voltage

Rotate the polarizer/analyzer pair so that your reference line makes an angle of 45° with the axes of the polarizers. Turn on the chopper, and record the transmitted light with the detector and the oscilloscope as shown in Figure 1. Now connect the reversible high-voltage DC power supply to the modulator input, and record the intensity of the output light as a function of applied voltage. Do not apply more than 1000V to the modulator. Suggested voltages to apply are 0, 50, 100, 150, etc...

Now turn off the power supply, reverse the polarity of the input and repeat the experiment. Record your data in tabular form in your notebook (similar to the table shown below). Plot the results graphically and paste your graph into the your notebook.

Voltage	Intensity (Normal Polarity)	Intensity (Reverse Polarity)
0		
50		

100		
150		
200		
250		
300		
350		
400		
450		
500		
550		
600		
650		
700		
750		
800		
850		
900		
950		

5.2.4 (a) Half-Wave and Quarter-Wave Voltages

From your graph, estimate the quarter-wave voltage $V_p/2$ and the half-wave voltage V_p . Compare your measured halfwave voltage with your calculated result from (5.21). What can you say about the "natural bias" of the modulator? Also estimate the differential drive voltage, which will give a reasonably linear response.

5.2.4 (b) Shifting the Modulator Zero-Voltage Bias Point

1. As you will note from your plot in (5.23), the modulator output intensity as measured between crossed polarizers is non-zero when the applied voltage is zero. This is because the natural birefringence and the fixed length of the crystal combine to yield output light that is in general elliptically polarized. If we were lucky, the output light exiting the crystal would have been either linear or circularly polarized (two special cases of elliptically polarized light). However, the devil is always at work, and we are not lucky today. To minimize the zero-voltage output intensity, a wave plate of the proper phase retardation would have to be inserted between the crystal and the analyzer to convert the elliptical polarization back to linear or circular polarization. Based on your data, what is the required static phase retardation, G_0 , of the desired wave plate for (a) linear polarization output, and (b) circular polarization output?
2. We have only two wave plates in the laboratory: one quarter-wave and one half-wave plate. The best we can do is to place one of these plates between the modulator and the analyzer and adjust its angular position along with that of the analyzer to achieve the desired overall modulator bias point. Which plate do you think is the most versatile for this task? Try them both to verify your answer. First turn off the DC power supply and disconnect it from the modulator. Remove the chopper from the output beam. Now place the half-wave plate between the crystal and the analyzer and do a systematic search of all combinations of half-wave plate orientation and analyzer orientation to achieve the best possible extinction of the beam. Repeat the process with the quarter-wave plate. Which one works better? Convince the TA or LA of your answer with a demonstration. Explain in your lab notebook why the plate you have chosen works better and why the other plate is not as effective?

5.2.5 Frequency Response of Modulator/Drive Electronics System

Connect the high-voltage audio amplifier to the modulator. Apply a sinusoidal modulating voltage to the crystal (using the low-voltage signal generator to drive the amplifier) and observe both the drive signal waveform and the modulated light waveform on the oscilloscope. First we can find a linear operating region for the system by placing one of the wave plates between the crystal and the analyzer, and adjusting both the wave plate and the analyzer so that the output optical signal amplitude as seen on the oscilloscope is optimized (undistorted, in phase with the driving voltage, and as large in amplitude as possible). Obviously we should use as small an input signal as necessary when performing this polarization optimization step so as not to overdrive the modulator. The input signal amplitude may be increased after the optimization is performed.

Measure the amplitude of the drive signal as well as the intensity of the transmitted light as a function of frequency. Use 1, 10, 102, 103, 104, 105 and 106 Hz for input frequencies. Record your data in tabular form in your notebook. Use this information to plot the frequency response of the system and paste your plots into your notebook.

Also answer these questions: (a) Is the result what one would reasonably expect for an electro-optic modulator? Be sure to take the frequency response of the detector into account; (b) what do you think is limiting the frequency response of the system, and why?

5.2.6 Optical Communications Demonstration

Turn off and disconnect the low-voltage signal generator. With the system still optimized for upmodulation in the linear operating region, connect the electrical output of the cassette tape recorder, CD player, shielded computer cable or radio to the high-voltage amplifier and complete the setup of the optical communications link using the audio amplifier to drive the loud speaker (see Figure 1). Convince the TA or LA that your system works as a free-space optical communication system (Also see A. Yariv, "Optical Electronics in Modern Communications", Oxford, chapter 9). Comment on the performance of the system in your notebook.

5.3. Acousto-optic Modulation

The setup we will use for observing acousto-optic light modulation of a He-Ne laser beam is shown in Figure 5.

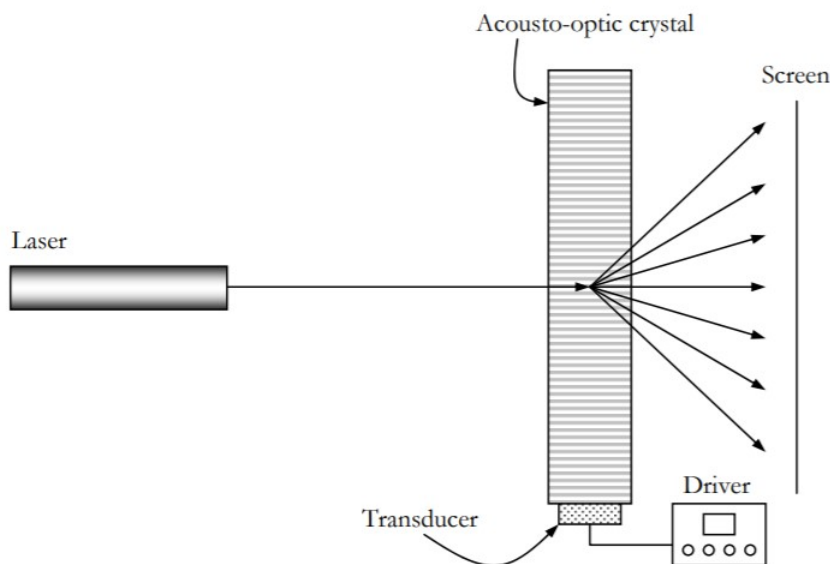


Figure 5. Set-up for studying acousto-optic light modulation

5.3.1 Measurement of the Grating Period and Acoustic Velocity

1. With the center frequency control of the IntraAction DE-40M VCO (FM) driver set at about 3/4 of its maximum position, measure the frequency of this drive signal on the oscilloscope. When using the oscilloscope, be sure to turn the RF power level to its minimum position and use a 10x attenuator so as not to damage the scope. Next, measure the angular displacement of the first-order diffracted beam about 2 meters away from the deflector and record your data in your notebook. The center frequency range of DE-40M is about 30-60 MHz. Use this measurement to calculate the spatial frequency of the "grating". (That is, the

periodicity of the sound wave in the crystal.) Show your calculation in your notebook. Why is it not necessary to know the refractive index of the crystal for this calculation? Show your reasoning.

2. Use your data to calculate the speed of sound, v_a , in the crystal. By comparison with known material parameters, as may be found, for example, in Yariv's book, what are some possible materials that the deflector may be made of?

5.3.2 Diffraction Efficiency versus Drive Power and Intensity Modulation

1. With the setup still operating at the above-chosen frequency, vary the strength of the drive signal with the "RF level control" knob and note how the intensity of the first-order diffracted beam is modulated. Now plot the dependence of the relative power in the diffracted first-order beam (relative to the beam incident on the modulator) as a function of the drive power for about four or five values. Explain your observations and results in your notebook. Is the dependence what one expects from the theory?
2. Now keeping the amplitude of the drive signal constant, vary the drive frequency with the "centering" control and re-measure the first-order diffraction efficiency. Plot your results graphically for about six frequencies two of which are the minimum and maximum frequencies of the driver. Why does the diffraction efficiency of the first order fall off with frequency?

5.3.3 Frequency Modulation

The DE-40M VCO can also convert amplitude modulated electrical signals to frequency modulated electrical signals which then drive the deflector. Connect the low-voltage signal generator to the VCO analog input, and for frequencies less than 10 Hz observe the diffracted beams on the screen. Do not put more than 1 volt into the DE-40M oscillator. Use a detector and a knife edge (razor blade) in the path of the output beam (appropriately expanded) to convert the spatially oscillating first-order beam to an amplitude modulated electrical output (see Figure 6).

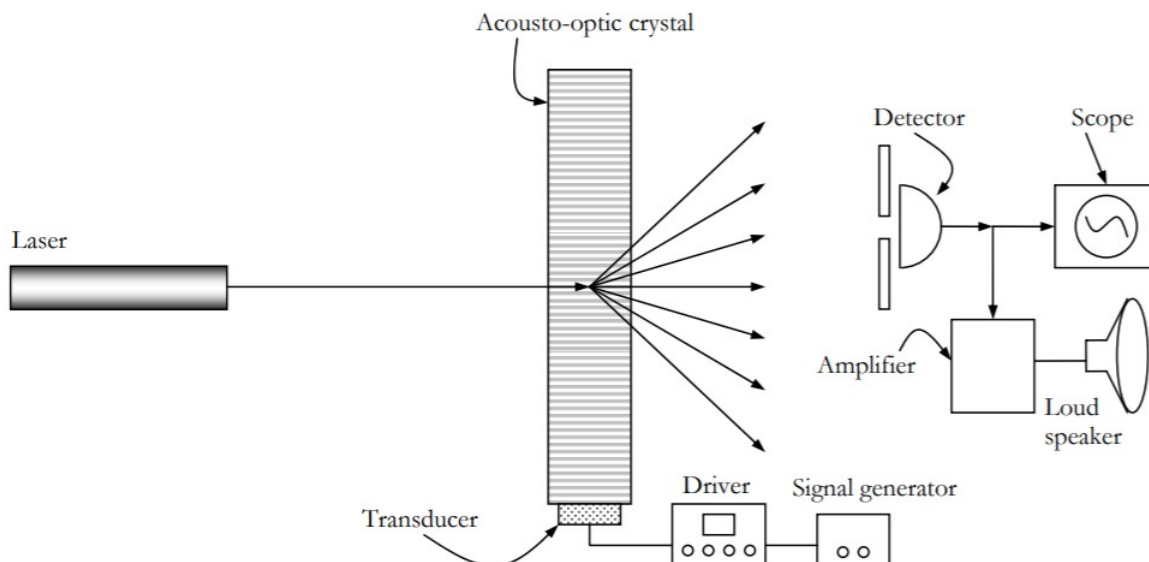


Figure 6. Setup to demonstrate frequency modulation and optical communications

5.3.4 Optical Communications Demonstration (optional)

Use a CD-player as the input to the DE-40M, and adjust or modify the system in Fig. 6 to make it function as a simple acousto-optic communications system. Convince the TA or LA that your system works as a free-space optical communication system. Comment on the quality of the reproduction.

Following the thinking you did on the prelab, now build the optical system you designed in the pre-lab exercise and place it in the system. Again, comment on the quality of reproduction. *Hint: Convert the FM signal to an AM signal with the use of lenses, mirrors, diffusers and/or spatial filters. Also, make sure that what you hear out of the detector is indeed the optic signal – ensure that you are not broadcasting from the nearby audio source via the audio cable (ensure that the detector and speaker are in a within a Faraday shield).*

5.4 Real-Time Holography in Photorefractive Crystals

Objective

The goal of these experiments is to investigate real-time holographic information recording, storage, and readout in two different photorefractive recording media: a barium titanate (BaTiO_3) crystal and a bismuth silicon oxide ($\text{Bi}_{12}\text{SiO}_{20}$) crystal. The experiments are, for the most part, qualitative in nature. A full interpretation of the observations will require familiarity with charge transport in semiconductors, and the tensorial properties of the electro-optic effect.

The Crystals

Three crystals are provided: A, B (both barium titanate, BaTiO_3) and C is a bismuth silicon oxide crystal ($\text{Bi}_{12}\text{SiO}_{20}$). Crystal A has a roughly square cross-section and all six sides are polished. Crystal A is to be used for the experiment 5.11 and 5.5. Crystal B has a rectangular cross-section and only two sides are polished. Crystal B is to be used for this experiment (5.4).

5.4.1 Hologram Recording, Storage, Readout and Erasure in BaTiO_3

In this experiment, a green argon laser beam ($\lambda = 514 \text{ nm}$) is split into two portions and then the two beams are recombined in the BaTiO_3 crystal so as to write a phase grating (the blue line of the argon laser at 488 nm could also be used, but it is less efficient than the green line for this experiment). A halfangle θ of about 10° between the two beams is typical. The grating is read out at the Bragg angle with a weak He-Ne laser beam ($\lambda = 633 \text{ nm}$) and the diffracted beam intensity is recorded with the aid of a photodetector. The set up is as shown in Figure 7.

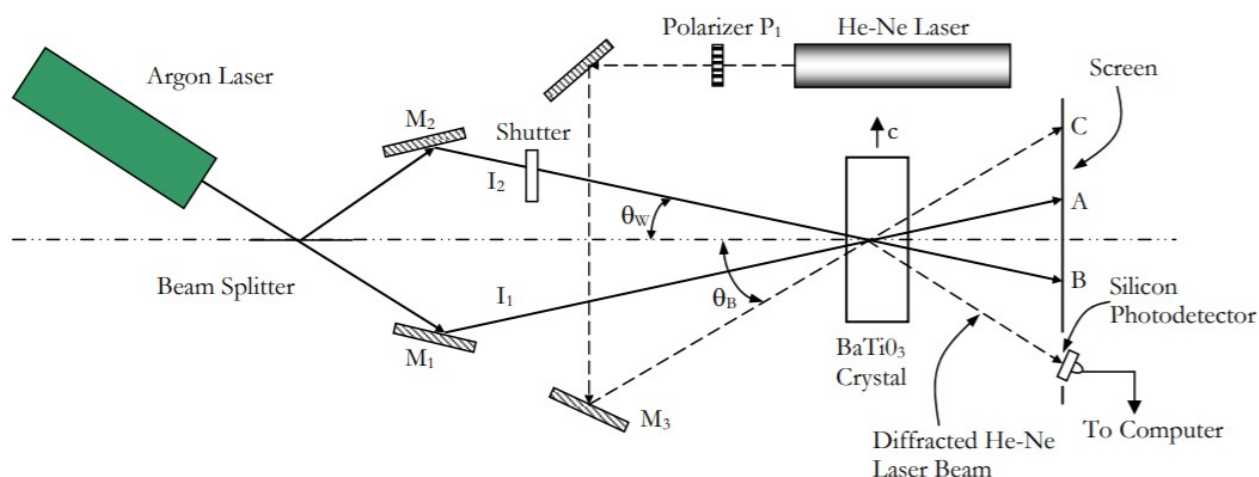


Figure 7. Set-up for measuring the diffraction efficiency of a BaTiO_3 crystal.

You should find the system already set up with the c-axis of the crystal in the plane of incidence of the argon laser beams as shown above. The angle of incidence of the He-Ne laser beam has been preadjusted so that the Bragg condition is satisfied. A fiber-optic coupled detector placed in the path of the diffracted He-Ne beam in conjunction with the BNC 210 National Instruments connector block and the computer is used to monitor the diffracted beam (alternatively, one could use a large-area photodetector and mask off the active area with a piece of black tape or cardboard to leave only a small opening roughly the size of the diffracted beam). The polarizer P1 in front of the unpolarized He-Ne laser serves to alter the polarization of the red He-Ne readout laser beam incident on the crystal.

Rotate the polarizer so that the He-Ne laser light is polarized in the plane of incidence. Place the screen in front of the photodetector, so that the transmitted Argon laser beams, as well as the transmitted and diffracted He-Ne Laser beam can be seen on the screen.

To make sure that the system is functioning properly, first erase the grating in the crystal. This is accomplished by blocking one of the arms of the argon-laser-beam-writing system using the electronic shutter so that the crystal is flooded uniformly with the light

from the second arm for about 60 seconds. The intensity of the diffracted He-Ne Laser beam on the screen should slowly disappear. Now open the shutter so as to record a phase grating in the crystal. The diffracted intensity on the screen should rise quickly to its saturation level.

5.4.2 Bragg Readout

1. With respect to the BaTiO₃ crystal, is the He-Ne laser beam an ordinary beam or an extraordinary beam?
2. Measure the polarization of the Argon laser beam. Is it ordinary or extraordinary polarized?
3. By measuring the actual separation between the transmitted Ar laser beams on the screen, determine the angle between the two writing beams. (Do not touch the crystal when making these measurements).
4. Calculate the period of the sinusoidal refractive index modulation that exists in the crystal using the results of part (3).
5. Calculate the Bragg angle for He-Ne laser readout from your result in part (4). Compare this result with the measured Bragg angle, and explain any discrepancy between the two results.
6. To experimentally find the Bragg angle, θ_B , we first calculated it to get an approximate value. Then an incremental search was performed about the calculated value (by changing the angle of incidence of the He-Ne laser beam on the crystal) until the maximum diffracted He-Ne beam intensity was obtained. Ask the TA or LA to demonstrate how extremely strict the conditions are for Bragg matching.

5.4.3 Diffraction Efficiency

1. Remove the screen and use the Newport digital power meter and associated detector to measure the intensity of the He-Ne laser beam. With P1 adjusted so that the readout light (He-Ne laser) is polarized within the plane of incidence (which contains the c-axis of the crystal) open the shutter and observe the buildup of the diffracted light on the photodetector. It is very important that you not bump or otherwise disturb the setup while a grating is being written into the crystal. Write down the saturation intensity level of the He-Ne diffracted beam. Now place the Newport power meter in the readout He-Ne beam (before it enters the crystal) and measure the full intensity of the beam. Calculate the diffraction efficiency (power in the diffracted beam divided by power in the incident beam) of the crystal under these illumination conditions.
2. Record the relative intensity of the argon laser beam using the Newport digital power meter.
3. Now replace the photodetector in the path of the diffracted beam and connect the photodetector output to the computer. Use software provided to record the data on the digital chart-recorder (in LabView). Erase the crystal, then open the shutter and record the build-up of the diffracted light as a function of time on the digital chart recorder. Paste your data, or a reduced version thereof, into your report. Record the time scale on your plot, and calibrate the vertical axis using the saturation diffraction efficiency level of Part (a) as a benchmark. Identify the 10-to-90 percent rise time and the peak value of the diffracted He-Ne laser light intensity. Record these values on your plot.

5.4.4 Erase Time

With the same configuration as in 5.4.3, write the grating until the diffraction efficiency saturates. Now, close the shutter so as to block the light in one of the arms of the writing beam, and let the other argon laser beam, I₂, readout the crystal. Record the decay of the grating by monitoring the diffracted argon laser beam intensity on the digital chart recorder as I₂ erases the grating while reading it out. Place your data, or a reduced version thereof, into your report. What is the 90-to-10 percent erase time?

5.4.5 Storage Time

With the same configuration as in 5.4.3, write the grating until the diffraction efficiency saturates. Now block the argon laser beam just after it exits the argon laser so that no write light reaches the crystal. Note the storage time of the grating, by setting the chart recorder for a 7-minute sweep and record the decay of the grating (as read out by the He-Ne laser) over the 7-minute period. Paste your data into your report. From your, data estimate the storage time of the crystal under these particular readout conditions.

5.4.6 Intensity Dependence of Write and Erase Times

Repeat the experiments in 5.4.3 and 5.4.5 with the argon laser beam at 3/4 of its original power. The laser power-control knob is on the back of the laser. Get the TA's help in adjusting the power of the laser beam. Paste your data into your report. Using your data, comment on the dependence of the diffraction efficiency, write time and the erase time on the pump power (argon laser power).

5.4.7 Dependence of Diffraction Efficiency on the Polarization of the Readout Beam

Repeat experiment 5.4.3 once more with the readout He-Ne laser light polarized perpendicular to the plane of incidence. Notice that the results are different from the original experiment with the He-Ne laser polarized in the plane of incidence. The point here is that

it is possible to create a refractive-index grating (phase grating) that cannot be "seen" by readout beams of the "wrong" polarization. This happens frequently in anisotropic dielectric media where D_n is a function of the polarization of the light reading out the crystal.

5.4.8 Self-Pumped Wavefront Phase Conjugation

Barium titanate is unusual among photorefractive crystals in that it exhibits the phenomenon of selfpumped phase conjugation. That is, a single laser beam entering a barium titanate crystal at the appropriate angle is retroreflected out of the crystal and back along its original path with the same wavefront shape as the incident beam (see Figure 8). Thus, the crystal acts as a phase-mirror.

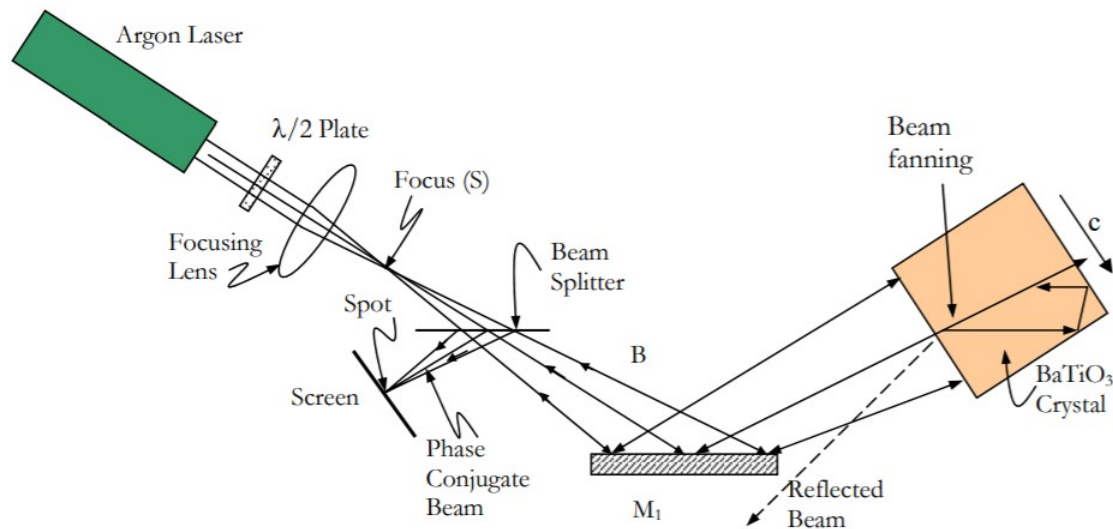


Figure 8. Illustrating the arrangement used to demonstrate self-pumped phase conjugation.

Beam fanning and two-beam coupling are essential to the mechanism involved in the self-pumped phaseconjugate process. These effects are stronger in BaTiO₃ for extraordinary polarized light (polarization along the c axis) than for ordinary polarized light.

To explore the phase conjugate property of the barium titanate crystal, we will make the following changes to the setup of Figure 7.

- Use crystal A. Place the crystal on the stand so that the chip on the long edge is at the bottom, the chipped corners are on the back, and the reflected beams from the front face of the crystal make an angle of about 30° with the incoming beam.
 - Change the wavelength of the argon laser light to the blue line at 488 nm (TA will do this), and
 - Modify the set-up of Figure 7 to that of Figure 8 without the lens. This is accomplished by closing the shutter in Figure 7 so as to get a single beam, and using the given halfwave plate to rotate the polarization of the argon laser beam into the plane of incidence (which contains the caxis of the crystal). The key feature of this setup is a single extraordinary polarized writing beam interacting with the crystal.
1. With the approximately-collimated laser beam entering one of the square faces of the crystal, rotate and translate the crystal as necessary until the phase-conjugate beam appears on the screen. You will have to wait about two to three minutes before the phase-conjugate beam is visible. Beam fanning within the crystal (see Figure 8) should be visible as the phase-conjugate beam builds up. Note that the phase-conjugate beam is approximately collimated even one meter away from the setup. Convince the TA or LA that you have observed the phase conjugate beam.
 2. A second optional demonstration involves placing a weakly focusing lens (one that does not significantly overfill the crystal) in the argon laser beam as shown in Figure 8. The focused spot S before the beam splitter serves as the input-object for this demonstration. The phaseconjugate beam should appear as a focused spot of light at the same distance from the beam splitter as S is in front the beam splitter. Move the screen about one meter away from the setup and note that the phase-conjugate beam is divergent, as expected.
 3. A third optional experiment involves removing the lens from the system, and placing the given aberrator in region B of the beam path. Describe and explain your observations.

4. Remove the aberrator after steady state has been reached. Describe and explain the response due to the removal of the aberrator.

5.4.9 Photo-refraction in Bi12SiO20 (optional)

Bismuth Silicon Oxide is known to have cubic symmetry. Repeat all of the above experiments on bismuth silicon oxide. Compare your results with those for the BaTiO3 crystal and explain the differences in each case. What is your estimate of the relative photorefractive writing speed of Bi12SiO20 vs BaTiO3 for the same write laser power?

5.4.10 Summary (optional)

Given your results and observations in the experiments above in addition to all the other reading you have done concerning the photorefractive effect, and in particular the manifestation of this effect in BaTiO3, write three or four paragraphs summarizing the highlights of this fascinating phenomenon that would be exciting to a typical layman who would read a journal at the level of Scientific American.

APPENDIX

Computer operating instructions

VI folder

Advanced data logger

Advanced data logger Device 1 (computer channel)

Digits of precision: 5

Channel: 0

Run

Save

Enter header text

(Use matlab script to read VI file.)

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CHAPTER OVERVIEW

6: Solid-State and Gas Lasers

[6.1: Pre-lab Exercises](#)

[6.2: In-lab Exercises](#)

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6.1: Pre-lab Exercises

Solid-State and Gas Lasers

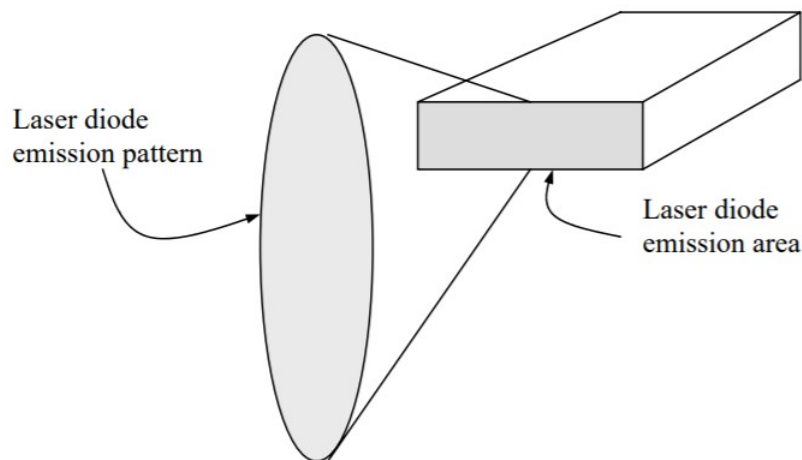
The Pre-Lab Exercises must be completed BEFORE entering the Lab. In your lab notebook record data, explain phenomena you observe, and answer the questions asked. Remember to answer all questions in your lab notebook in a neat and orderly fashion. No data are to be taken on these laboratory sheets. Tables provided herein are simply examples of how to record data into your laboratory notebooks. Expect the in-lab portion of this exercise to take about 3 hours.

PL 6.1 – Get Prepared to Start the Laboratory Exercises

Read the entire laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. **Write** them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade.

PL 6.2

Design a lens system that would convert the beam from a typical diode laser to a circular shape as is done in laser pointers. Remember that diode lasers typically have an emission area with a rectangular cross section. The beam exiting a laser diode can be described by two spatially independent Gaussians (in x and y). The total beam is the result of the multiplication of these two Gaussians. One Gaussian will have much greater divergence than the other (so the intensity pattern will look like an oval, instead of a circle).



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6.2: In-lab Exercises

Solid-State and Gas Lasers

6.1 Specific Laser Systems

1. In this exercise, we want you to examine and identify the key components within a variety of commercial and homemade laser systems. These include:

- He-Ne laser
- Argon-ion laser
- Homemade CO₂ laser
- Nd:YAG laser
- Visible semiconductor diode laser
- Infrared semiconductor diode laser

Specifically, you should look for things such as, output wavelength, length of the cavity, type of mirrors used, Brewster windows, pumping mechanism, typical out power, typical input electrical power, wall-to-light efficiency, mode locking or Q-switching mechanisms (describe their operating principles), expected coherence length, and cost.

For each of the six lasers, write brief comments on each of the characteristics listed in the above paragraph. As an example, you would fill in the missing information in the table below with text or numbers as appropriate for the chemical laser. Hint: go to the manufacturer's website, [Photonics Buyer's Guide](#), or [Laser Focus Buyers Guide](#).

Chemical Laser

- *Operating wavelength:*
- *Approximate Cavity length*
- *Approximate length of gain medium:*
- *Type of mirrors used:*
- *Type/configuration of Brewster windows:*
- *Pumping mechanism:*
- *Typical optical output power:*
- *Electrical power requirements:*
- *Wall to light efficiency:*
- *Mode-locking mechanism:*
- *Q-switching mechanism:*
- *Expected coherence length:*
- *Approximate cost:*
- *Manufacturer:*

2. For the open-cavity He-Ne laser, observe the light emitted from the side of the laser tube with the spectroscope. Describe your observations and their implications in terms of what you know about lasers and about the He-Ne laser system in particular. Add your description and analysis to your table in (a) for the He-Ne laser.

6.2 Semiconductor Diode Laser Characterization

1. Using the Tektronix 576 curve tracer the Hitachi HL 6320G 10mW visible laser, connect the purple and white leads to the collector and emitter slots of the curve tracer. Make sure to set the base terminal to "open." Also, set the vertical scale to 200mA/div and set the horizontal scale to 500 mW/div. The laser has a peak wavelength of 632 nm with maximum and minimum wavelengths of 640 nm and 625 nm respectively.
 - Record the i-v characteristics of the laser diode.
 - Direct the output light from the laser onto the silicon photodetector. Slowly increase the current into the laser diode, being careful not to saturate the detector or destroy the laser diode, and plot the output intensity-vs.-current characteristic for the laser.

Estimate the threshold current from your plot.

2. Using the HL 6320G semiconductor diode laser, record a picture of the output beam shape using a diffuser and the digital camera.

3. Measure the divergence angles of the output beam.
4. Use your measurements of Part (b) to estimate the cross-sectional dimensions of the laser cavity.
5. Coherence Length Measurement: The setup shown below is a Michelson Interferometer that can be excited simultaneously by both a He-Ne laser and the diode laser whose coherence length we want to measure. In this experiment, both laser beams are collimated before they reach the interferometer. As the He-Ne laser has a much longer coherence length than the diode laser, the main purpose of the He-Ne laser is to establish interferometer alignment at all times.
 - With the two arms of the interferometer unequal by about 20 cm, align the system so that a spatial fringe pattern is formed on the screen by the He-Ne laser. Make a sketch of and comment on the fringe pattern you have produced (perfect alignment is not necessary). Now block the light from the He-Ne laser so it no longer excites the interferometer. With the diode laser turned on, observe its spatial interference pattern (if any) on the screen. Make a sketch of and comment on the fringe pattern owing to the diode laser. Compare the two interference patterns - comment on their similarities and differences. Record the positions of the two mirrors of the interferometer.
 - Translate the movable mirror to new position, realign the system with the He-Ne laser beam, and repeat part (i).
 - Repeat part (ii) as necessary until you have enough information to get a rough estimate of the coherence length of the diode laser. Show you data for each position you tried.
 - Describe how you arrived at your estimate of the coherence length.

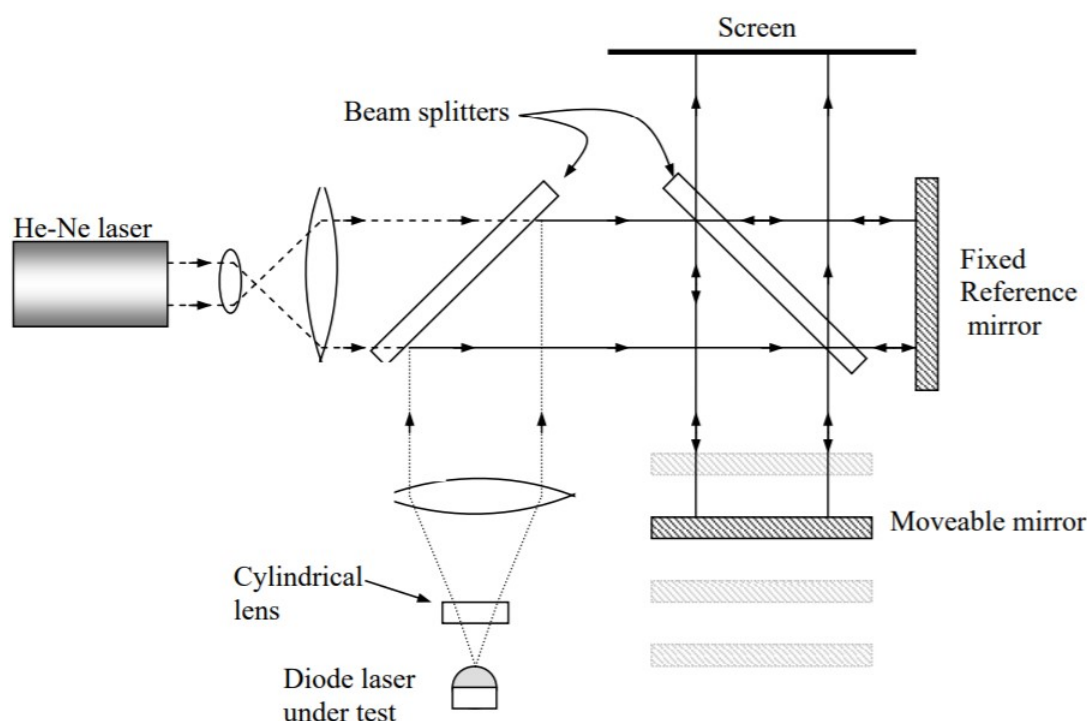


Fig. 1. Michelson interferometer setup to measure the coherence length of a diode laser.

6.3 Spectral Analysis of the HeNe Laser

1. The small, 2mW, He-Ne lasers in the modern optics lab are NOT monochromatic. Unfortunately the 1/2m Jarrel Ash grating spectrometer in laboratory does not have sufficient resolution to resolve the fine structure of the laser output. Although we know the He-Ne laser employs curved mirrors, for convenience in this exercise we will model its resonator as a Fabry-Perot

interferometer (see Fig. 2). The laser resonator (cavity) modes (high Q) naturally select the output frequencies for the laser.

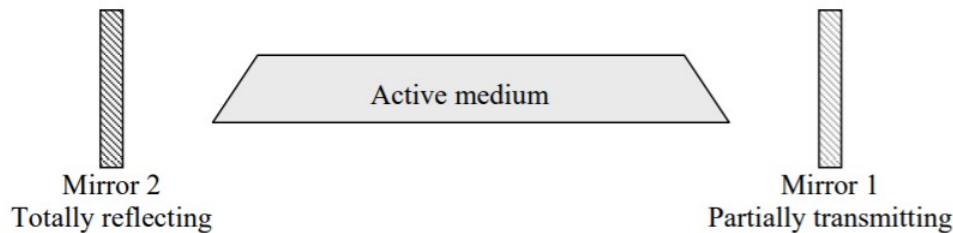


Fig. 2. Fabry-Perot Laser cavity

Lasing action thus occurs under the gain curve, $g(v)$, only at a discrete set of frequencies. However, in order to sustain laser oscillation, the gain per round trip in the cavity must be greater than the loss, and so no laser action occurs at Fabry-Perot transmission bands, which fall near the wings of the gain curve. The laser output therefore looks like that shown in the Fig. 3. These are the so-called longitudinal modes of the laser.

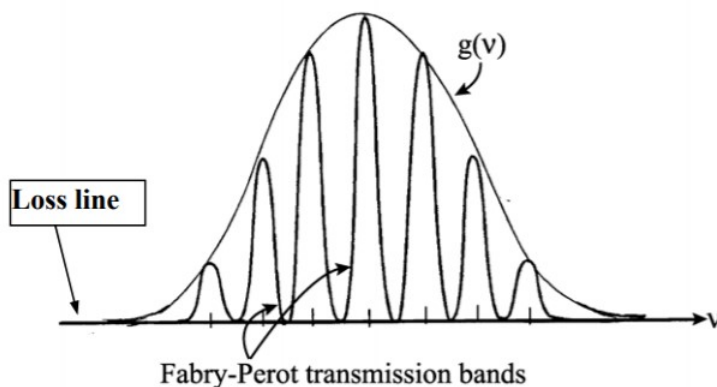
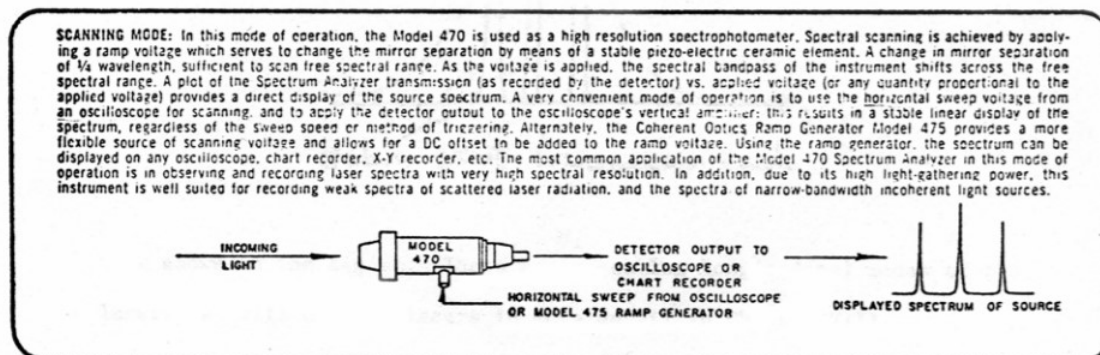


Fig. 3. Illustrating the possible oscillation frequencies of a laser

The scanning Fabry-Perot spectrometer is an instrument that can resolve the fine structure of the laser output. Its principle of operation is described in Chapter 3 of the Class Notes. You will be using this instrument to study the light output of our small He-Ne lasers. It is also called an optical spectrum analyzer.

The MOL has an “Optical Spectrum Analyzer” made by Spectra Physics Corp. Although it is actually a spherical Fabry-Perot interferometer in the confocal mirror arrangement we will approximate its operation by assuming a parallel-plate Fabry-Perot configuration. The free spectral range of the etalon is given to be equal to 2 GHz. A description of the scanning mode in which the device operates is given below:



The output of our 2-mW He-Ne laser consists of a number of closely spaced “modes.” Given that the gain bandwidth of the lasing medium (assume it is equal to the “Doppler width”) is approximately 1500MHz, how many modes would you expect to be present in the laser output? (See Fig. 4).

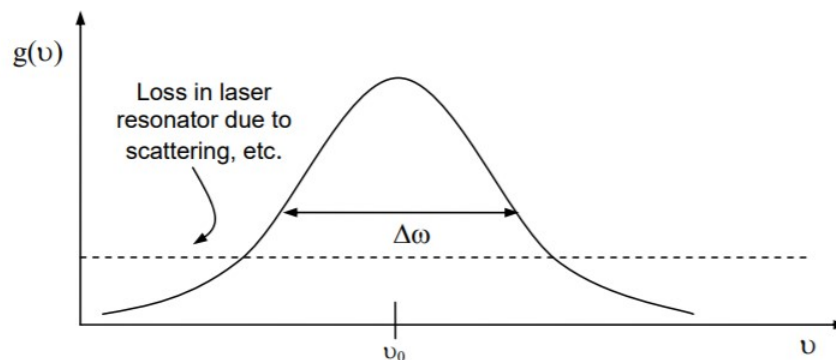


Fig. 4. Inhomogeneously broadened (Doppler broadened) He-Ne laser gain profile. Assume $\Delta\nu \approx 1500\text{MHz}$ and $\nu_0 @ 4.74 \times 10^{14}$ where ν_0 is the optical carrier frequency.

2. Measure the modes separation and compare your results with the estimate in part (a). How many free-spectral ranges (each equal to 2 GHz) does the scan of the interferometer cover? What is the separation between the two mirrors in the scanning Fabry-Perot interferometer?
3. Increase the dispersion by expanding the horizontal scale of the oscilloscope trace and measure the half-width of a single peak. Since the bandwidth of a single mode from the laser is much beyond the resolution of the Fabry-Perot, the measured width gives directly the resolution of the instrument. Calculate the finesse, F , for the instrument.
4. Observe the effect of placing a linear polarizer in the unpolarized laser beam on the mode structure. Note the results for various orientations of the polarizer. Try to explain the observed behavior.
5. Adjust the micrometers of the Fabry-Perot angular mount so as to direct reflected light from the interferometer back INTO the He-Ne laser. What would you expect? (Good alignment is critical).

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CHAPTER OVERVIEW

7: Optical Waveguides and Fibers

[7.1: Pre-lab Exercises](#)

[7.2: In-lab Exercises](#)

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7.1: Pre-lab Exercises

Optical Wave-Guides and Fibers

The Pre-Lab Exercises must be completed BEFORE entering the Lab. In your lab notebook record data, explain phenomena you observe, and answer the questions asked. Remember to answer all questions in your lab notebook in a neat and orderly fashion. No data are to be taken on these laboratory sheets. Tables provided herein are simply examples of how to record data into your laboratory notebooks. Expect the in-lab portion of this exercise to take about 3 hours.

PL 7.1 - Get Prepared to Start the Laboratory Exercises

Read the entire laboratory handout, and be prepared to answer questions before, during and after the lab session. Determine all the equations and constants that may be needed in order to perform all the laboratory exercises. **Write** them all down in your laboratory notebook before entering the Lab. This will ensure that you take all necessary data while in the Lab in order to complete the lab write-up. This preparatory work will also count toward your Lab Exercise grade.

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7.2: In-lab Exercises

Optical Wave-Guides and Fibers

Basic Fiber Handling and Measurements

The objective of these experiments is to familiarize the student with techniques for handling fibers, coupling light into and extracting light from fibers, and basic fiber-optic measurement techniques.

7.1 Examination of Various Fiber Types and Fiber-Cabling Technologies

Take a look at the single-mode, multi-mode and plastic fibers in the laboratory. Pay particular attention to their construction and the size of the core/cladding structure of each fiber with the aid of the microscope. Also look at the various types of connectors and connectorized fibers. Postulate as to the use of the various cabling members and applications for particular connector types.

7.2 Cleaving, Cleaning, Furcating, Connectorizing and Splicing Fiber

Because most optical fiber is very thin and very fragile, fibers can be difficult to use unless they are placed in a positioning ferule or connectorized or both. Positioning ferules and connectors allow a fiber to be coupled to a photo-detector or a light source. Splicing joins two fibers, sometimes through the use of connectors. All three operations involve stripping, cleaving, and cleaning at some stage (or several stages) of the process.

Basic Procedures for Stripping Fiber

"Bare" fiber is actually composed of three layers of material: the core, the cladding, and a protective buffer. The buffer is analogous to the plastic jacket on conductive wires. At any junction in the fiber (i.e., any point at which the fiber will be spliced or connected), the buffer must be removed. We use one of four tools to strip fiber in the lab. In each case, the buffer is stripped no more than a centimeter at a time.

1. **Wire strippers:** Operate the wire strippers just as you would use them to strip wire, while making sure that you do not stress the fiber (by pulling or bending) enough to crack the glass. Glass fiber is quite resilient to longitudinal strain, having a high tensile strength; however, glass fiber will easily snap when shearing stresses or bending (impact) stresses are applied.
2. **No-Nicks Fiber Stripper:** Assure that there is no residue in the blades of this stripper. Place the fiber between the triangular teeth of the tool with the fiber end in the direction of the arrow on the tool head. GENTLY release the grips and allow the head to close, ensuring that the fiber remains in place. Pull the tool head in the direction of the arrow to strip the jacket.
3. **Fiber stripping tool:** Assure that there is no residue in the blades of the stripper. Insert the fiber into the alignment hole at the front of the stripper until the end lines up at the desired point on the stripper scale (on the tool's handle). Squeeze the handles fully together and pull along the fiber axis.
4. **Chemical strippers:** On rare occasions you may need to use a chemical stripper as to not damage the cladding of the fiber. To do this, you will use paint thinner. By dipping the fiber in paint thinner for a period of 5 minutes, you will dissolve the fiber buffer. After dipping, any excess residue can be removed easily with a damp Kim wipe – be sure to avoid skin-contact with the paint thinner. We prefer not to use this method since paint thinner is toxic and creates strong fumes within the laboratory. Also, additional measures must be taken to dispose of old or used paint thinner with the environmental protection office at MIT.

Basic Procedures for Cleaning Fiber

Once the jacket has been stripped from a fiber, it is necessary to clean any remaining residue from the cladding. Simply fold a small Kim wipe into eighths and wet it with alcohol. Fold the wipe again, and insert the fiber into the "channel." Apply gentle pressure and pull the wipe toward the end of the fiber. Repeat until the fiber feels smooth while dragging the wipe.

Basic Procedure for Cleaving Fiber

The ends of a fiber must have flat surfaces (i.e., perpendicular to the fiber axis) to allow a good connection. Cleaving allows the precise termination of the end of a fiber. It always involves a process of first scoring the fiber and then breaking the glass along the scored line. We use two tools for cleaving in the lab:

1. **Cleaving with the scribe:** The scribe is a diamond-tipped tool that can be used to score the fiber. The first step while using the scribe is to secure the end of the fiber to the lab table. A lab partner or a piece of tape comes in handy for this exercise. Then, while pulling the fiber taut, gently drag the scribe tip across the fiber, perpendicular to its axis. Place a drop of water on the

scribed region of the fiber to facilitate the cleave. Finally, while applying pressure to the secured fiber end, pull the fiber straight back until it cleaves.

2. **Cleaving with a fiber-cleaving tool:** For a more accurate cleave, the T&B cleave tool should be used. Adjust the knob on the tool's side until the scale reflects the amount of stripped fiber that will remain. Then, insert the fiber into the screw end of the tool until it stops. Assure that the fiber exits at the gripping end of the tool. Close the latch at the screw end and squeeze the gripping end of the tool. Pull down the safety latch, and squeeze the scribe block. Release the scribe block. Pull the gripping end away from the scribe block to cleave the fiber.

Typically, the fiber is inspected after the cleaving process. The cleaved end should be flat across the fiber and perpendicular to its axis. There should be no tag protruding from the fiber, and the initial scribe should be less than 5% of the core diameter.

Basic Procedure for Connectorizing Fiber

Connectorizing gives a fiber a protected, polished end that is much more versatile and stable than an ordinary bare fiber. The operation for fiber connectorization with the _____ connector is as follows:

1. Clean the connectors that will be used. Any dust or broken fiber ends in the connector will make it difficult to apply glue to the fiber. Thus, soak the connector in isopropyl alcohol for a few minutes. Shake out any excess, and allow the remaining alcohol to evaporate. Inspect the ferrule (white ceramic tube) of the connector by holding it up to a light. Light should be visible through the connector body 30cm to 45cm from your eye. Occasionally, if the connector has been used before, a fiber end may be stuck in the capillary (hole) of the ferrule. If this is the case, use a small piece of piano wire to push the glass through the ferrule, and wipe and soak the connector.
2. Cut the furcation tubing to a length 3cm longer than the desired length. Use the Kevlar scissors to cut through the yellow strands.
3. Slide the strain relief boots and crimp sleeves onto the tubing.
4. Cut the outer tubing ~1.5cm from the end, leaving the Kevlar strands and any inner tubing intact.
5. Insert the fiber into the inner tubing and push through until the end of the fiber comes out the end of the tubing. Allow ~2.5cm of extra fiber on each end of the tubing.
6. Strip ~1.5cm of jacket from the fiber ends to be connectorized, and clean the fiber with isopropyl alcohol.
7. Carefully test fit the connector onto the stripped fiber end. Slowly rotate the connector to ease the fiber's entry into the ferrule. Pay close attention to the fiber: if it is bending under pressure, it is not entering the ferrule tip. Pull back, and attempt to find the capillary. Once the fiber end comes out the ferrule tip, continue to feed it through until it stops. Assure that at least 2.5mm of fiber exits the tip. Once the fit has been tested, remove the connector.
8. Prepare the epoxy packet by removing the separator bar and mixing the two components. Thread a needle onto the syringe, and remove the plunger. Remove the corner of the epoxy packet and fill the syringe with the contents. Invert the syringe, insert the plunger, and wait for the epoxy to clear from the syringe tip. Then, press the plunger to expel air from the tip. If necessary, tap the tip of the syringe to flush any air bubbles from the epoxy. (Air bubbles remaining in the cured epoxy will weaken the bond between the connector and the fiber.) Wipe any excess epoxy from the syringe needle.
9. Insert the needle into the back of the connector until it bottoms out. Maintain pressure between the connector and the needle, and press the plunger of the syringe to inject epoxy into the ferrule. Continue applying epoxy until a small bead appears on the outside face of the connector ferrule. Remove the needle.
10. Insert the fiber into the connector as before. Assure that the epoxy bead still remains to support the fiber. If the bead has diminished, it is possible to add epoxy directly to the connector tip.
11. Seat the connector. Slide the crimp ring over the end of the connector.
12. Using the 0.178" hex section of the crimp tool die, crimp the sleeve onto the connector. Then, using the 0.128" hex section, crimp the sleeve to the fiber furcation tubing.
13. Apply some epoxy to the inside of the strain relief boot, and slide it into place ensuring that the connector ring can still turn as necessary. Allow the epoxy to cure. Note: Curing can be accomplished by heating the connectors at 65 °C for 15 minutes and cooling for 15 minutes. Otherwise, allow the connectors to cure overnight.

Polishing Fiber Connectors

The final step in connectorizing a fiber is polishing. Polishing ensures that the fiber and ferrule tip is flat, smooth and clean.

1. Using a fiber scribe, score the fiber just above the epoxy bead, and cleave it. Ideally, the remaining fiber tip should extend no more than one fiber diameter above the epoxy bead.

2. Take a 1"x9" strip of a 5-micron lapping sheet and fold it in half without creasing the film. Gently polish the fiber tip until it is flush with the epoxy bead.
3. Clean the rubber polishing pad with isopropyl alcohol and a Kimwipe. Apply a 5-micron lapping sheet to the pad, rough side up.
4. Clean the polishing disc and the connector with isopropyl alcohol and a Kimwipe. Place the polishing disc on the lapping sheet. Gently place the connector in the hole of the disc.
5. Without applying pressure downward on the connector, begin polishing the fiber in a figure-8 pattern. If the fiber is contacting the lapping sheet, a light figure 8 pattern should appear on the polishing film. If no pattern appears, apply gentle pressure on the connector. Increase the pressure gradually as the epoxy bead is polished down. Inspect the connector end frequently, and continue polishing until a thin film of epoxy remains.
6. Repeat steps 3, 4, and 5 with the 3-micron lapping sheet until only a faint hint of epoxy remains on the connector end.
7. Repeat steps 3, 4, and 5 with the 1-micron lapping sheet, applying 3 to 4 drops of water on the polishing film. Continue until no epoxy remains on the connector.
8. Inspect the fiber tip with the fiber microscope. If necessary, clean the 1-micron sheet and polish the tip again.
9. Polish the tip with the 0.3-micron lapping sheet, once again using 3 to 4 drops of water.
10. Inspect the fiber tip for the following:
 - The connector end surface should be free of epoxy.
 - The fiber should be flush with the end of the connector ferrule.
 - There should be no heavy scratches through the core of the fiber.
 - There may be light random scratches, but there should not be any chips in the edges of the fiber.

Whenever using connectorized fiber, make sure that the ends are clean. Use isopropyl alcohol and a Kimwipe to make sure the surface is dust and oil free.

7.3 Estimation of Numerical Aperture (NA) of a Multimode Fiber.

The numerical aperture of a fiber is defined as the sine of the maximum acceptance half angle of the fiber. In order to get a reasonable measurement of numerical aperture it may be necessary to first perform mode stripping on the fiber, especially if a short length of fiber is being used. Secondly, the output end of the fiber must be polished or cleaved so it is flat. Thirdly, the fiber must be overfilled with light. This is, the cone angle of the light coupled into the core of the fiber must exceed the acceptance angle of the fiber – we can do this through the use of a high-power microscope objective.

The setup for the measurement of numerical aperture of a multimode fiber is shown in Fig. 1. In this system, since fluctuations in the laser light can affect the measurement, a fraction of the light from the He-Ne laser can be split off using the beam splitter and monitored on a reference detector R. The major portion of the light is transmitted through the beam splitter and coupled into the fiber with the aid of a microscope objective X and a fiber micropositioner M. The output light from the end of the fiber, which is held stationary, can be viewed on a screen S or a photodetector D which sits on a rotation stage calibrated in degrees. To get rid of higher-order modes traveling in the cladding, we could use a sleeve that surrounds the cladding with mode-stripping fluid (see Fig. 1) – however omitting this device will not significantly affect the results for this exercise.

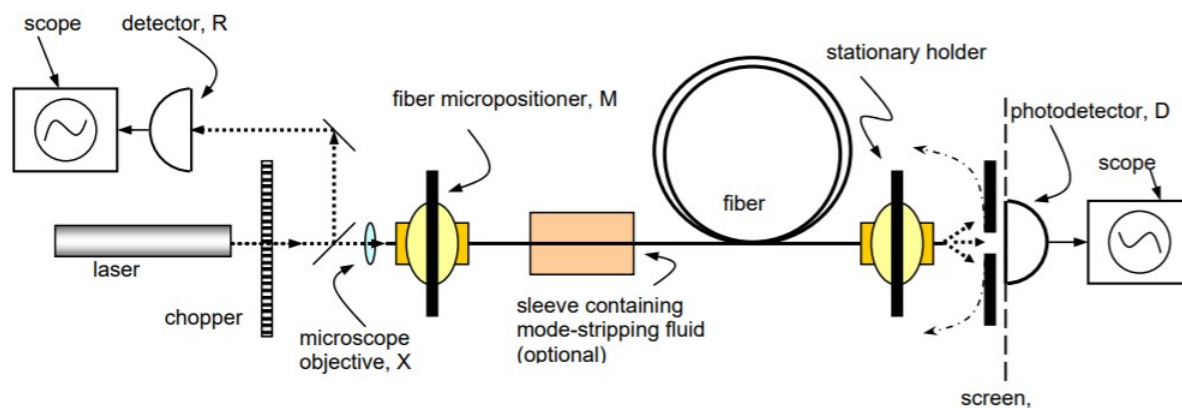


Figure 1. Set-up for measurement of numerical aperture.

First make a visual observation of the numerical aperture by placing the small white screen a few centimeters away from the output end of the fiber and observing the size of the spot on the screen. By making rough measurements of the spot diameter and the distance between the end of the fiber and the screen, a crude estimate of the numerical aperture can be obtained.

1. For a more precise measurement of the numerical aperture the detector and the calibrated rotating stage are used. The end of the fiber is secured to a post a few centimeters above the axis of rotation of the table and at the same level as the detector. With the fiber stationary, the base of the stage is rotated so the detector sweeps out an arc around the end of the fiber. Measurements of the detector power should be made every two degrees along with the corresponding value on the reference detector.

The ratio of the output light intensity to the reference light intensity should then be plotted as a function of angle, and the numerical aperture determined from the plot. These curves could be repeated for launch conditions corresponding to an underfilled condition to show that, in this case, the measured result underestimates the numerical aperture of the fiber. Record the data and calculate the NA in your laboratory notebook.

Note on mode stripping.

Mode stripping is the process of removing high-order modes propagating in the cladding of a fiber. Mode stripping is often necessary when making measurements on fibers to avoid erroneous results. Such is the case for the numerical aperture measurement in Fig. 1 above. Mode strippers are generally placed near the input and output ends of a fiber, so that those highorder cladding modes which result from overfilling at the input end are removed along with loworder modes that convert to high-order modes in the cladding because of scattering along the length of the fiber. For a bare glass fiber mode stripping can be accomplished by passing the fiber through a sleeve filled with honey, glycerin or mode-stripping index-matching fluid, which provides an approximate index match with the cladding.

1. Using the fiber manipulator and the microscope objective, couple light from the laser into one of the fibers and perform mode stripping near the input end using the honey or glycerin jacket as shown in Fig. 1. Observe the evanescent losses through the honey or glycerin.
2. Bend the fiber and observe the decrease in output signal on the scope as well as the light leakage from the fiber (a high curvature will be needed, but please do not break the fiber).

7.4 Mode-Distribution and Polarization Properties of Single-Mode Fibers

Observation of modes in a single-mode fiber

Couple the He-Ne laser beam into the single-mode (Coreguide or SMF-28) fiber and observe the mode patterns of the few low-order modes. Note that the fiber is not single mode at the He-Ne laser wavelength of 633 nm. To get a clean pattern, the ends of the fiber must be terminated properly. Sketch the patterns you see in your notebook – then identify which modes you have drawn (they might be composite modes – mixtures of more than one type of mode pattern).

Observe the polarization characteristics of these low-order modes by placing a polarizer in the path of the output beams. Write your observations in your laboratory notebook.

Polarization properties of single-mode fiber

In general, when an optical fiber is excited with linearly polarized light, the light exiting the fiber is no longer linearly polarized. By twisting a single-mode fiber with a rotation about the fiber axis, it is possible to alter the polarization state of the light in the region of the twist because strain-induced birefringence in the glass.

The “rabbit ears,” a type of polarization controller, are designed to apply such a twist to the fiber. Using this apparatus in conjunction with two linear polarizers, apply a twist to the fiber and monitor the polarization state of the light exiting the fiber. Try to obtain a linearly polarized output beam without breaking the fiber.

7.5 Observation of Bending and Splicing-Misalignment Losses (optional)

Couple He-Ne laser light into the multi-mode fiber that is threaded through the “rabbit-ears”, and observe the leakage of light out of the fiber especially in the small radial bend on the inner loop of the “rabbit-ears”. Write your observations in your lab notebook.

Next couple the light from the output of this fiber into a second fiber by but coupling the ends of the two fibers. Observe the output of the second fiber on a screen. Investigate the coupling losses for axial offset, angular offset and separation as illustrated below.

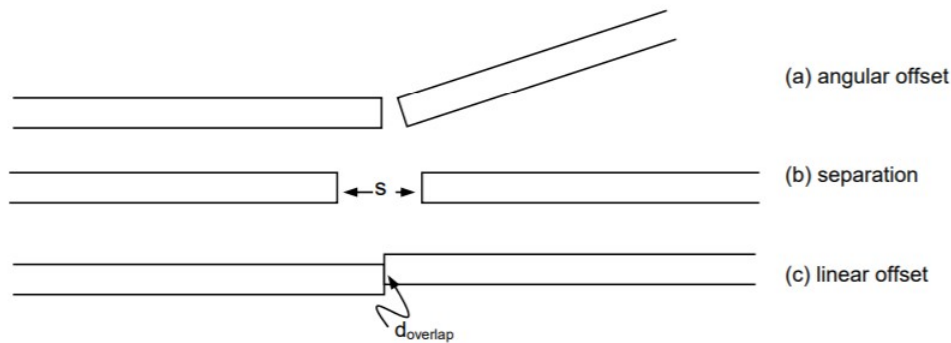


Figure 2. Types of Splicing Losses

Now, determine the loss in decibels incurred by connecting two FC-PC-type connectors together. Do the same for the loss incurred by using a fiber splice. Which technique for connecting two length of fiber together is better? Why? What determines if one method is for connecting two fiber lengths together is 'better' than another?

7.6 Fiber Attenuation Coefficient of Inter-Office Multimode Plastic Fiber Cables (Optional)

For this equipment (cable substitution method) you will use the following equipment:

1.) FOTEC "S" consisting of a variable frequency pulse generator driving an LED at 665nm or similar connectorized pulsed source.
 2. FOTEC "C" optical power meter
 3. Oscilloscope
 4. Three lengths of plastic fiber l1, (short) and l2 (long), and an unknown length l3.
- Connect the FOTEC "S" to the FOTEC "C" with a short length of cable. Observe, using the dual trace oscilloscope, the internal electrical drive signal to the FOTEC "S" LED, and the output of the FOTEC "C" detector by connecting the outputs of both these instruments to the oscilloscope. What is the frequency range of the FOTEC "S"?
 - Choose a reasonable operating frequency for the pair of instruments (for good optical power measurements) and use the cable substitution method described in the notes to measure the attenuation coefficient of the cables.
 - Insert an unknown length l3 of cable, and measure the power transmitted through the unknown length. Use this result and the attenuation coefficient of the fiber to estimate the length l3.
 - Comment on the practical limitations of the above method of measuring attenuation coefficient.

7.7 Imaging Properties of a Coherent Guided-Wave Bundle

A stack of thin cover glass slides has been provided (see Fig. 3). Observe the imaging properties of this stack by using it to read text on a printed page. For example, what do you see when you move your head back and forth in the x direction? In the y direction? Why? Explain its principles of operation. How does this compare to what you would see had you used a solid block of glass with the same shape? Or bundle of optical fibers with the same shape? Why?

Also provided is a naturally-occurring rock (Ulexite), which has waveguiding properties. Repeat the above experiment with the Ulexite sample. Observe the crisp high-resolution imaging properties of the rock. Postulate as to how the rock works (you may also do some outside research to find out how this rock is created in nature – or how synthetic versions are created in the laboratory).

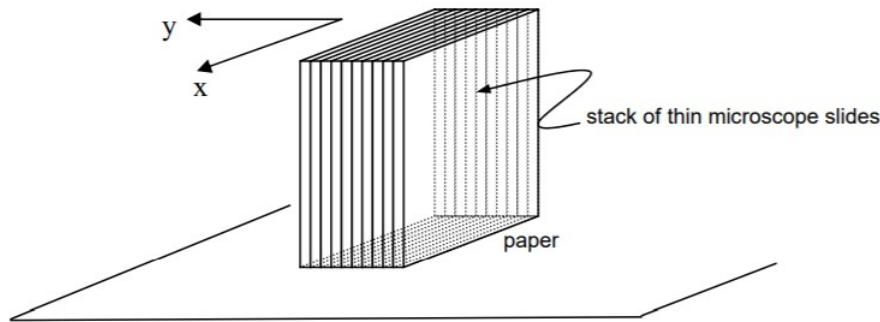


Figure 3. Imaging properties of a coherent stack of thin microscope slides

7.8 Wavelength Multiplexing Wavelength Multiplexing, Group Index and Dispersion

In this experiment, (see Fig. 4) you are given two Fiber Optic Stable Sources (FOSSs), each of which is internally or externally modulatable. One of the sources operates at a wavelength of 1310nm and the other operates at 1550 nm. You are also provided with two loops of connectorized fiber, and several FC-connectorized patch cables, all with an effective group index of $n = 1.468$. Additionally, you are given two Newport benchtop Wavelength Division Multiplexers (WDMs) and two Thorlabs high-speed InGaAs fiber optic photodetectors. You are also provided with a digital signal generator, numerous equal-lengths of BNC cable, and an oscilloscope.

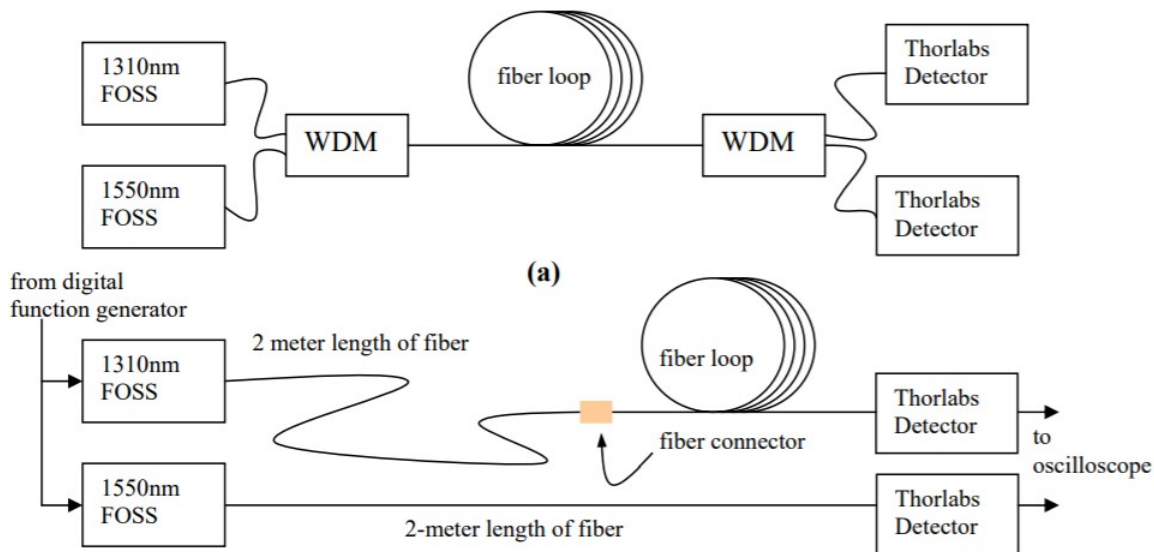


Figure 4. (a) WDM system for communications demonstration (b) system for measuring refractive group index of a single-mode fiber

Here the goal is to generate two optical signals, both infrared, and then wavelength division multiplex them into an optical fiber. Both signals will carry a digital waveform. At the output end of the fiber the signals are to be demultiplexed and separated, and the input signals recovered.

1. Try to measure the signal to noise ratio of the system, as determined by the cross-talk between the two input optical signals at the output of the system. Modulate the sources with their internal frequency driver circuitry and view the detector output on the oscilloscope.
2. By modulating one of the sources with a high-frequency external digital waveform, determine the length of the 900 mm furcated spool. **Make sure the input modulation signal goes no higher than 4 volts peak-to-peak or you will destroy the fiber optic source!**

3. Now, given that the un-furcated fiber spool has a length of 100 meters, determine the group index, n , for that fiber.
4. Operating with both sources at a high external modulation rate, and with both fiber spools connected between the WDMs, can you determine the difference in arrival times between the 1310nm and 1550 nm light? Does this correlate what you have learned about dispersion in step-index single-mode optical fibers?

7.9 Pulse Broadening (optional)

Here the goal is to make a qualitative measurement of multimode pulse broadening as a function of fiber length and numerical aperture and fiber type.

1. Compare the pulse shape from the short fiber used earlier in the cable substitution experiment (7.6) with the shape after propagating through one of the long fibers. With the long length of fiber, change the microscope objective illuminating the fiber from low power to high power (to change the numerical aperture) and re-observe the pulse shape.
2. Compare the broadening in a step index fiber with the broadening in a graded fiber of approximately the same length, keeping the numerical aperture constant.

II. FIBER OPTIC SENSORS AND SYSTEMS (possible projects)

The goal of these projects is to familiarize students with various fiber-optic sensor technologies.

7.10 WDM Video-Rate Transmission System

You have already observed the WDM system in the previous part of this laboratory exercise. However, you will now look at it in terms of a communication system. You are given two modulator circuits used to convert the analog video input into a serial digital output. These signals are transmitted over a length of fiber optic cable by use of a WDM. The signals are then recovered and converted to an analog signal to be displayed on side-by-side video monitors.

7.11 Sagnac Fiber-Optic Gyroscope

7.12 Fiber-Optic Refractometer

7.13 Fiber-Optic Thermometer

7.14 Fiber-Optic Velocimeter

7.15 Fiber-Optic Endoscope

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